



KTU NOTES

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**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

MODULE 1

NEED FOR NTMM(Non Traditional Machining Methods)

The requirements that lead to the development of nontraditional machining are

1. Very high hardness and strength of the material. (Above 400 HB.)
2. The work piece is too flexible or slender to support the cutting or grinding forces.
3. The shape of the part is complex, such as internal and external profiles, or small diameter holes.
4. Surface finish or tolerance better than that obtainable conventional process.
5. Temperature rise or residual stress in the work piece is undesirable.

COMPARISON BETWEEN TRADITIONAL &NON TRADITIONAL MACHINING

Traditional	Non Traditional
Material removal by shearing of work piece material	Alternate energy sources can be used for material removal and takes place by melting &vaporisation,anodicdissolution,impact erosion etc.
Can be applied to materials depending upon the strength & hardness	Can be applied to all metals & alloys
Higher MRR	Lower MRR
Low power consumption	High Power consumption
Chip formation is at macroscopic level	Chips formed in the microscopic or atomic levels

Not possible to machine many newer materials developed for aerospace and nuclear applications	Can be used to machine such difficult – to-machine materials
Accuracy & surface finish is low	More accurate & high surface finish
Direct contact between tool and work piece and tool should be harder than work piece material	Not necessary for a tool. Even if the tool is present it need not be harder than work piece material

COMPARISON BETWEEN TRADITIONAL & MICRO MACHINING

Traditional	Micro Machining
Miniature features like micro sized holes cannot be machined.	Can machine thousands of holes having a few microns diameter.
Not possible to machine materials with extra ordinary properties.	Can be used to machine such materials having high strength, high heat resistance and high hardness.
Nano level surface finish on complex geometries are impossible to achieve	Nano level surface finish on complex geometries can be achieved on conducting or non conducting materials

POWDER METALLURGY

Powder metallurgy is a process of making components from metallic powders.

Initially, it was used to replace castings for metals which were difficult to melt because of high melting point.

The development of technique made it possible to produce a product economically, and today it occupies an important place in the field of metal process.

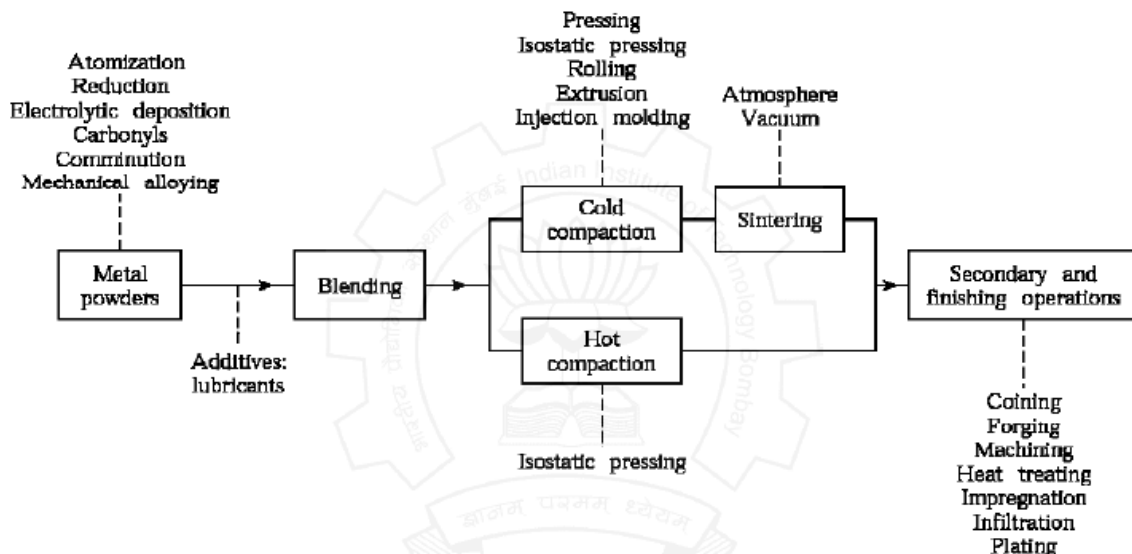
NEED OF P/M (Powder Metallurgy)

Initially powder metallurgy process was used to replace casting for metals which were difficult to melt because of their high melting point.

The development of techniques made it possible to produce the component economically, and today it occupied an important place in the field of metal processing. The number and variety of products made by powder metallurgy are continuously increasing; and includes the tungsten filament for lamps, contact point relays, self-lubricating bearings and cemented carbide for cutting tools.etc.

POWDER METALLURGY PRODUCTION STEPS

1. Manufacture of metal powders in suitable size & purity.
2. Blending & compacting of metal powders in suitable die cavity to provide Green strength.
3. Sintering the compact part at high temperature to provide bonding strength.
4. Secondary processes such as finishing, sizing & heat treatment



Steps in powder metallurgy

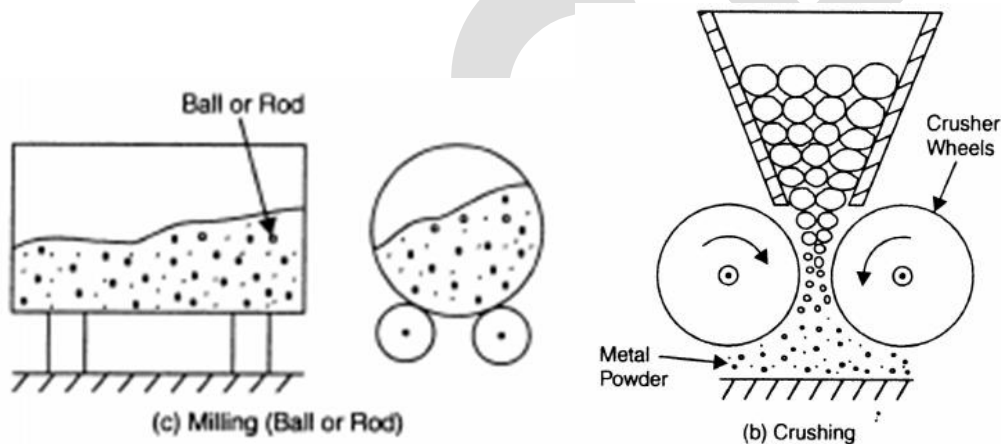
POWDER PRODUCTION METHODS

1. Mechanical pulverization

It includes machining, milling or grinding.

Machining: In this method first chips are produced by filing, turning etc. and subsequently pulverised by crushing and milling. The powders produced by this method are coarse in size and irregular in shape. Hence, this method is used for special cases such as production of magnesium powder.

Crushing and Milling: These methods are used for brittle materials. Jaw crushers, stamping mills, ball mills are used to breakdown the metals by crushing and impact.



2. **Shotting:** Dropping the molten metal through a sieve or small orifice in to water. This produces spherical particles of larger size. This method is commonly used for metals having low melting point.
3. **Electrolytic deposition:** Used for producing Cu & Fe powders. This process is similar to electroplating.

The powders have dendritic structure with low apparent density. This method consists of depositing metal on cathode plate by conventional electrolysis process. The cathode plates are removed and deposited powder is scrapped off. The powder is washed, dried and screened and oversized particles are milled or ground.

The powder is further subjected to heat treatment to remove work hardening effect. By this method powder produced will be pure and resist oxidation. The cost of manufacturing is high.

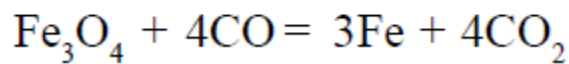
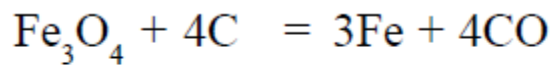
4. Chemical Reduction:

Pure metal is obtained by reducing its oxide with a suitable reducing gas at an elevated temperature (below the melting point) in a controlled furnace. The reduced product is then crushed and milled to a powder.

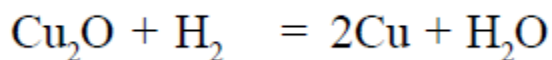
Irregular shaped powder is obtained.

Used for producing Fe,Cu,Ni,Co,Mo,W,Th,Ti

Iron powder is produced this way



Copper powder by

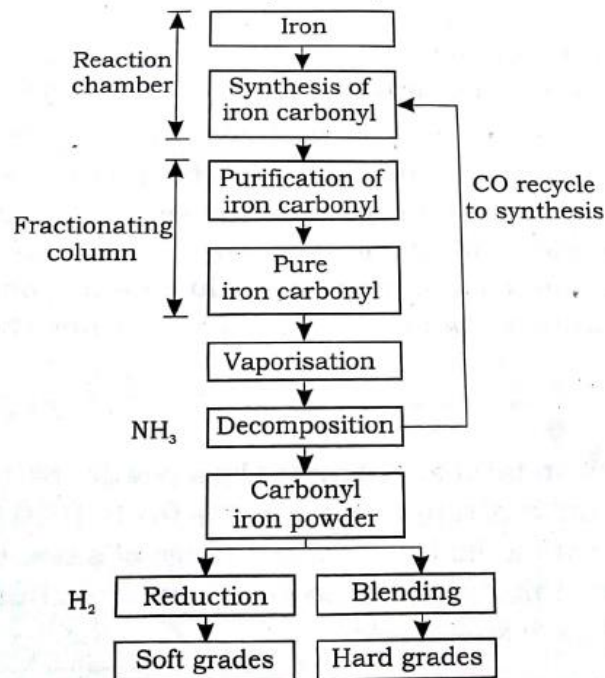


Tungsten, Molybdenum, Ni and Cobalt are made by the method

5. Reduction of carbonyls

This process is based upon the fact that a number of metals can react with carbon monoxide to form carbonyls such as iron carbonyl can be made by passing carbon monoxide over heated iron(75°C) at 50 – 200 bar pressure. The resulting carbonyl is then decomposed by heating it to a temperature of 200 – 3000°C yielding powder of high purity, however, at higher cost.

The carbonyl iron powder obtained after reduction is usually spherical & very fine.



6. Atomization:(Very Important)

This is the most popular and versatile method available for manufacturing metal powders. It can be used to make powders of most of the metals and alloys. Here, the molten metal is converted into a fine spray of droplets which solidifies into small particles to form a powder.

In this process the molten metal is forced through an orifice into a stream of high velocity air, steam or inert gas. This causes rapid cooling and disintegration into very fine powder particles and the use of this process is limited to metals with relatively low melting point.

Produces spherical particles with smooth surfaces

4 Methods of atomization(Very Important)

1. Gas Atomization

Figure 1 shows two different ways in which gas atomization can be implemented. Here, a high velocity stream of gas is used to convert the liquid metal into a fine spray. The air or inert gas is allowed to flow through an expansion nozzle, which siphons the liquid metal from the crucible below.

As the liquid metal comes up, it is carried away by the high pressure gas and is sprayed into a container as shown in figure 1. The spray process creates tiny droplets of metal which solidify into powder form.

A similar process is shown in figure 2. The molten metal reaches to the gas nozzle by way of gravity. The gas/air jets immediately atomise the liquid to form nearly spherical droplets of the liquid metal. The droplets cool down to solid powder particles which are collected in the chamber provided.

Used for HSS, Titanium, Super alloys (Inconel) & Reactive metals (Li, Na, K etc.)

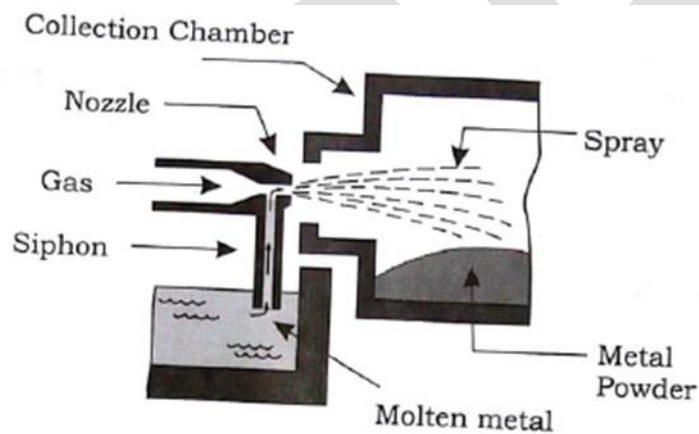


Fig.1

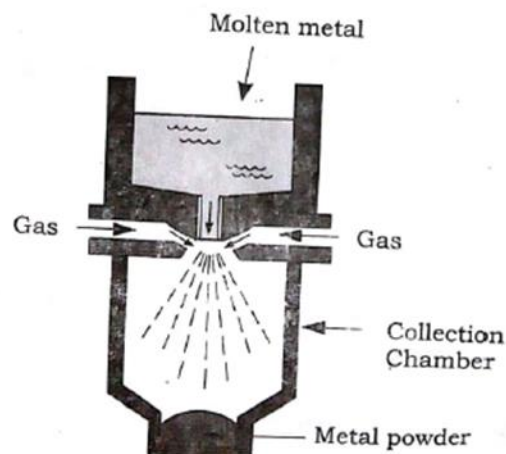


Fig.2

Process variables are jet distance, jet pressure, nozzle geometry, velocity of gas and temperature of melts.

Higher pressure or smaller jet distance produce finer powder.

The atomisation method produces spherical powder particles with relative smooth surfaces.

This method is suitable for HSS, Titanium, Super alloys (Inconel) & Reactive metals (Li, Na, K etc.)

2. Water Atomization

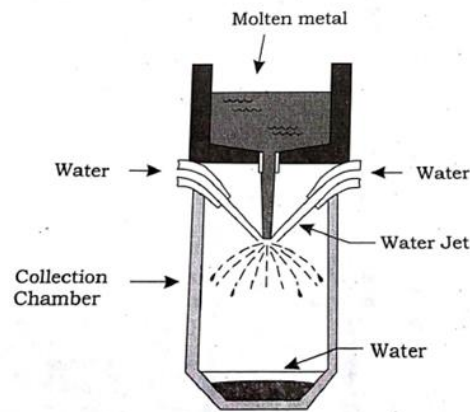
This process is very much similar to the gas atomisation except that a water jet is used instead of air jet.

Water atomization process is the most common of the atomisation methods, and is suitable for metals having melting point below 1600°C . As water is coming into direct contact with the hot liquid metal, the cooling is very fast but the powder obtained will be having irregular in shape.

The presence of water causes oxidation of the particle surface. To avoid this problem, water can be replaced with synthetic oil. This method is used for low and high alloy steels and stainless steel, but the particles are irregular in shape.

In these atomisation processes, particle size of powder is governed by the velocity of the fluid stream. When velocity of jet is increased, finer particles are obtained

This method is suitable for metals having MP below 1600°C . Used for LCS & SS. Synthetic oil can also be used.



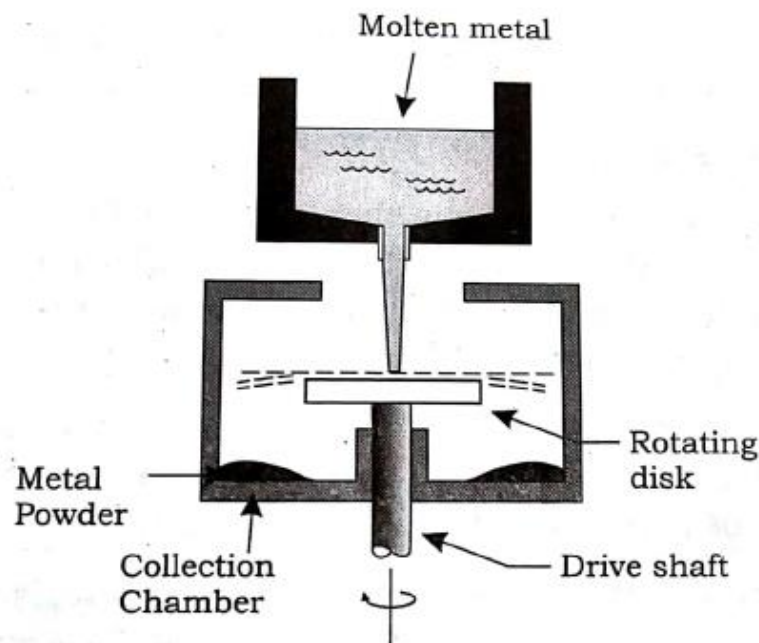
Water atomisation

3. Centrifugal Atomization

1) Rotating disk method

Rotating disk method is one of the processes under this category. The liquid metal is allowed to fall on a disk rotating at high velocity. Due to the centrifugal action, the liquid metal particles are thrown out in all directions.

The arrangement is shown in Figure. The powder particles are collected in the chamber surrounding the rotating disc.



Rotating disc method

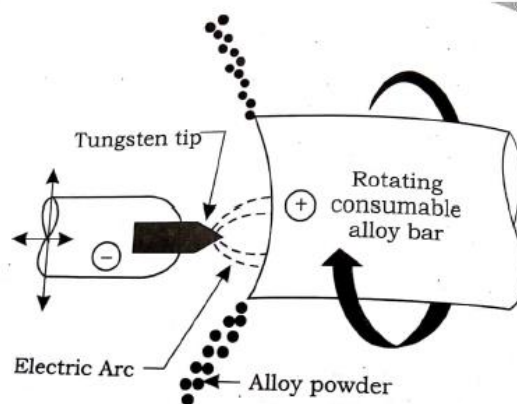
2) Rotating electrode method

Rotating electrode process (REP) is another example of centrifugal atomization. In this method, a solid metal bar is rotated at a faster rate, while one end is melted by an electric arc (as shown in figure).

The electric arc is established between a tungsten electrode (cathode) and the solid metal bar (anode). Molten metal spins off from the bar and solidifies immediately.

The electrodes are enclosed in an inert gas chamber and the powder particles are collected inside the chamber.

Accurate control of the rotation of anode is important to obtain a proper particle size distribution for the powder. Diameter of a droplet depends on surface tension of the liquid and the centrifugal force acting on it.



Rotating electrode method

4. Liquid gas Atomization

This process is of recent origin, in which a cryogenic liquid gas (argon-nitrogen) at -200°C is used to atomise the molten metal. The liquid gas kept at 300 bar pressure is used to generate an even stream which atomises melt and cools it rapidly. Upon atomisation, the liquid evaporates quickly and the powder is left behind cooled down.

Here the cooling rate is ten times higher than that of gas atomisation. The powder obtained is much purer than obtained from water atomisation.

Uses cryogenic liquid gas (Ar/N₂) at -200°C at 300 bar. Powder produced is more pure

CHARACTERISTICS OF METAL POWDER

- (a) Particle shape
- (b) Particle size
- (c) Particle size distribution
- (d) Flow rate
- (e) Compressibility
- (f) Apparent density
- (g) Purity

1. PARTICLE SHAPE:

The particle shape depends largely on the method of powder manufacture.

The shape may be spherical, irregular, angular, and dendritic.

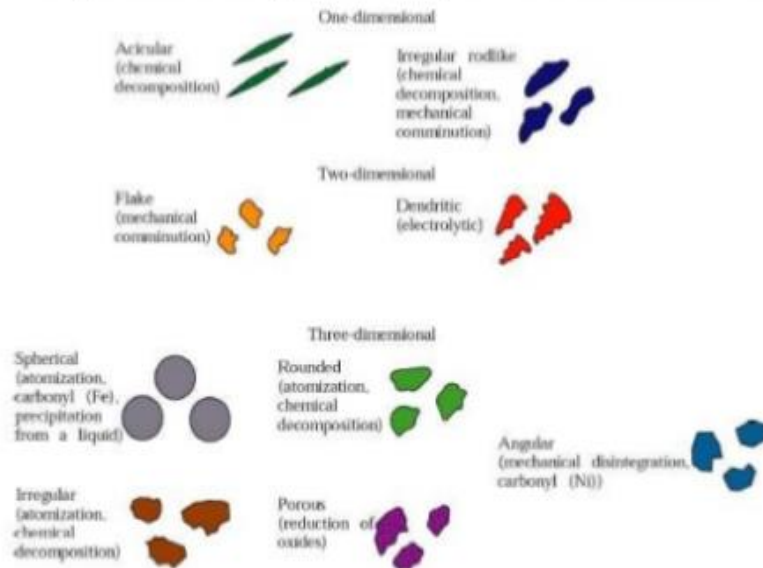
Among these spherical is ideal one which is having maximum apparent density and flow rate but reduced pressing and sintering capabilities.

The particle shape influences the flow characteristics of powders.

The rounded shape is observed in powders of Al, Cu, Sn & Zn obtained by atomization method.

Particle shape

- depends on powder production method



2. PARTICLE SIZE

It is defined as the total space occupied by a powder particle. It varies from 0.1 to 1000 microns. The size is obtained by passing the powder through standard sieve ranging from 45 to 150 μm mesh.

Particle size also controls the flow rate, compressibility, apparent density and sintering ability.

Fine mesh powder result in poor apparent density and flow rate and maximum sinterability.

Coarse powder gives good apparent density & flow rate but lower sinterability.

3. SIZE DISTRIBUTION

Each powder material has particles of different sizes, the sizes decreases continuously from a maximum size down to minimum size.

It is specified in term of a sieve analysis, the amount of powder passing through 100, 200 etc., mesh sieves. Particle size distribution influences the packing of powder and its behaviour during moulding and sintering.

4. **FLOWABILITY:** It is the ability of powder to flow readily and conform to the mould cavity. It determines the rate of production and economy.
5. **COMPRESSIBILITY:** It is defined as volume of initial powder (powder loosely filled in cavity) to the volume of compact part. It depends on particle size, distribution and shape.
6. **COMPACTABILITY:** Ability of powder to achieve a determined volume and weight at a given compacting pressure.
7. **APPARENT DENSITY:** It is the mass per unit volume of loose or unpacked powder. Apparent density depends on density of solid material, particle size, size distribution and shape of the particle.
8. **GREEN STRENGTH:** Refer to the strength of the compact part prior to sintering and it helps to handle the products during mass production. It depends on compressibility of metal powders.
9. **PURITY:** Metal powders should be free from impurities as the impurities reduces the life of dies and affect sintering process. The oxides and the gaseous impurities can be removed from the part during sintering by use of reducing atmosphere.
10. **SINTERING ABILITY:** It is the ability which promotes bonding of the particles by the application of heat.

BLENDING

The process of mixing powders is called blending.

A single powder may not fulfil all the requisite properties and hence, powders of different materials with wide range of mechanical properties are blended to form a final part.

Need for Blending

1. Blending imparts uniformity in the shapes of the powder particles,
2. Blending facilitates mixing of different powder particles to impart wide ranging physical and mechanical properties,

3. Lubricants can be added during the blending process to improve the flow characteristics of the powder particles reducing friction between particles and dies,
4. Binders can be added to the mixture of the powder particles to enhance the green strength during the powder compaction process.
5. Lubricants such as graphite powder, steric acid, zinc stearate are added in the blending operation to reduce die wall friction and to aid in the ejection.

COMPACTING

The loose powders are formed into shape by compacting or briquetting.

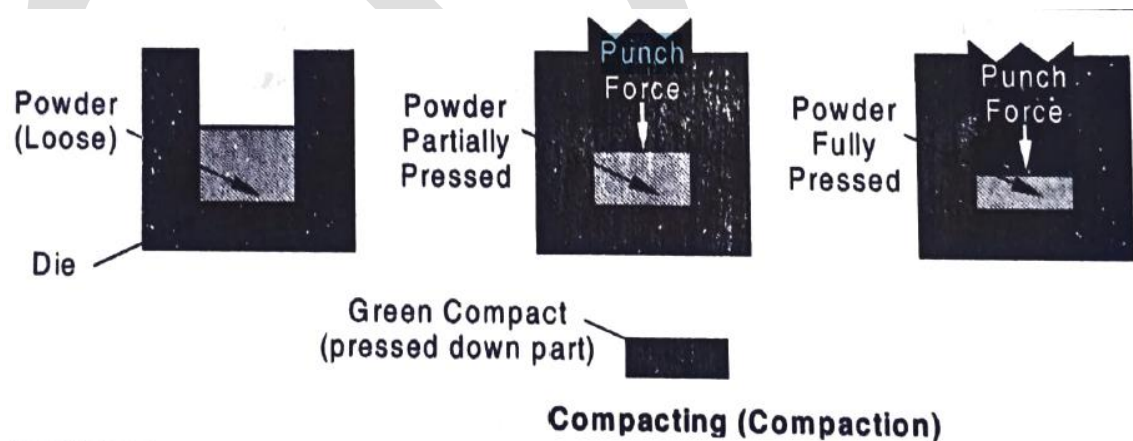
In this process loose powders are placed in a die and squeezed together to form a green compact having final shape and size of the component.

It is usually performed at room temperature.

Stages in compaction

In the first stage of powder compaction, density is increased by a rearranging of the individual powder particles.

Spaces, bridges and gaps are eliminated, and density increases due to a more efficient packing of the particles with applied pressure.



Effects of compaction

1. Reduces voids between the powder particles and enhance the density of the consolidated powder.
2. Produces bonding of the powder particles to improve green strength in the consolidated powder particles,
3. Facilitates plastic deformation of the powder particles to conform to the final desired shape of the part,
4. Enhances the contact area among the powder particles and facilitates the subsequent sintering process.

Green Strength means the strength of the compact part prior to sintering operation and helpful for handling of objects.

COMPACTING METHODS

1. Pressing
2. Centrifugal compacting
3. Slip casting
4. Extruding
5. Gravity sintering
6. Rolling
7. Isostatic pressing
8. Explosive compacting
9. Fibre metal process

1. PRESSING:

Metal powders are placed in a die cavity and compressed to form a component shaped to the contour of the die.

Pressure usually employed is 80 Mpa to 1400 Mpa. Light pressure is applied for soft metal powders used for porous bearings and high pressure for high density components from fine and hard metal powders. Mechanical presses are used for compacting objects at low pressure. Hydraulic presses are for compacting objects at high pressure.

2. CENTRIFUGAL COMPACTING:

The powder is twirled in a mould and packed uniformly with pressures up to 3 MPa. The uniform density is obtained for the component.

This method is employed for heavy metals such as tungsten carbide

3. SLIP CASTING:

In this method, the powder is converted into slurry with water and poured into the mould made of plaster of paris. The liquid in the slurry is gradually absorbed by the mould leaving the solid compact within the mould. The mould may be vibrated to increase the density of the compact. The main drawback of this process is relatively slower process because it takes larger time for the fluid to be absorbed by the method.

4. EXTRUSION

This method is employed to produce the components with high density. Both cold and hot extrusion processes are for compacting specific materials.

In **cold extrusion**, the metal powder is mixed with binder and this mixture is compressed into billet. The binder is removed before or during sintering. The billet is charged into a container and then forced through the die by means of ram. The cross-section of product depends on the opening of the die.

Cold extrusion process is used for cemented carbide drills & cutters.

In the **hot extrusion**, the powder is compacted into billet and is heated to extruding temperature in non-oxidising atmosphere. The billet is placed in the container and extruded through a die.

This method is used for refractory materials, Beryllium and nuclear solid materials.

5. GRAVITY SINTERING:

This process is used for making sheets for controlled porosity.

In this process the powder is poured on ceramic tray to form an uniform layer and is then sintered up to 48 hours in ammonia gas at high temperature. The sheets are then rolled to desired thickness.

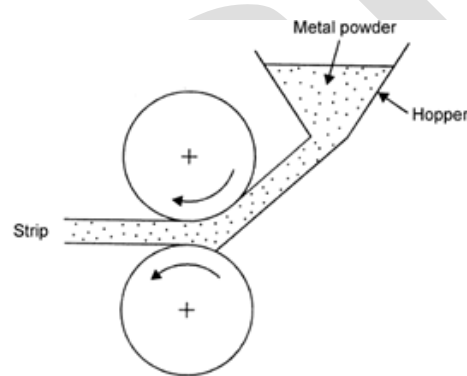
Porous sheet of stainless steel are made by this process and popularly used for filters.

6. ROLLING:

This method is used for making continuous strips and rods having controlled porosity with uniform mechanical properties.

In this method, the metal powder is fed between two rolls which compress and interlock the powder particles to form a sheet of sufficient strength. It then sintered rerolled and heat treated if necessary.

The metals that can be rolled are Cu, Brass, Bronze, Ni, Stainless steel and Monel.



7. ISOSTATIC PRESSING(Very important)

In this method, metal powder is placed in an elastic mould which is subjected to gas pressure in the range of 65-650 Mpa from all sides.

After pressing, the compact is removed from gas chamber. If the fluid is used as press medium then it is called as hydrostatic pressing.

The advantages of this method are: uniform strength in all directions, higher green compact strength and low equipment cost.

This method is used for tungsten, molybdenum and reactive materials such as beryllium, uranium, zirconium, titanium

2 methods of ISOSTATIC PRESSING(Very important)

- 1. CIP(Cold Isostatic Pressing)**
- 2. HIP(Hot Isostatic Pressing)**

1. CIP(Cold Isostatic Pressing)

In CIP the pressing is carried out at room temperature.

The metal powder is placed inside a flexible rubber mould and hydrostatic pressure is applied to it.

Water or oil is used as the working fluid.

Thus, the powder gets compacted and the green compact is further taken out and sintered.

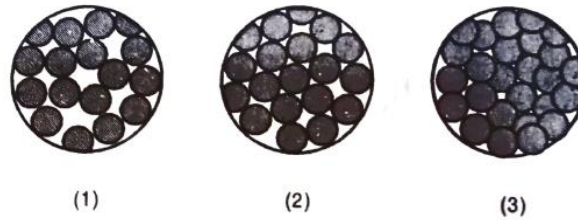
This is an economical process which leads to uniform density of compaction.

Since the mould is flexible dimensional accuracy cannot be achieved.

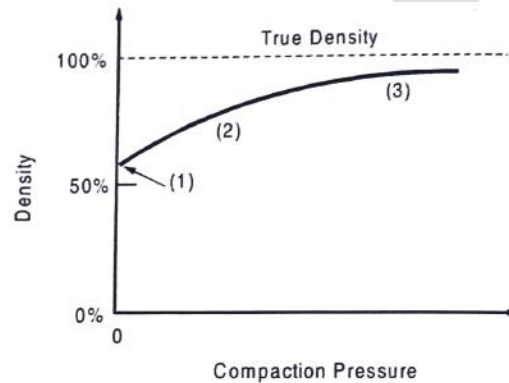
Pressure distribution in CIP

1. At first, the pressure applied in compaction results in repacking of the powders into an additional efficient arrangement, eliminating 'bridges' formed during filling, dropping pore space, and rising the number of contacting points among particles.
2. If the pressure is increased, the particles are plastically deformed, causing the interparticles make contact with area to boost and additional particles to make contact.
3. By further reduction in pore volume, this can be done.
4. The progression is illustrated in the following figures.

For P/M compaction, usually mechanical, hydraulic or combination of both mechanical and hydraulic presses are used.



(a) Effect of Applied Pressure during Compaction: (1) Initial Loose Powders after Filling, (2) Repacking, and (3) Deformation of Particles



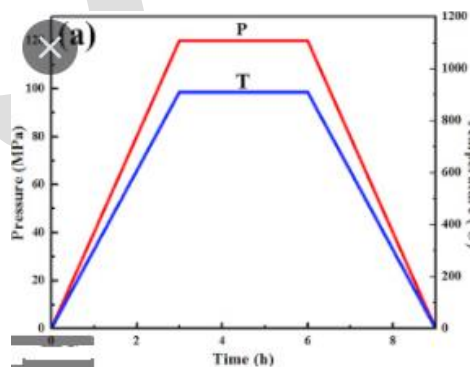
2. HIP(Hot Isostatic Pressing)

In this process, metal powder is stressed using inert gas in a metal container at pressure and temperature of 100 MPa and 1000°C respectively.

The container used in this process has very high melting point.

An inert gas is used as the pressing medium.

Here a metal powder is filled into a mold and then it is surrounded by a secondary pressing medium. After that, vacuum is created and the entire assembly is kept in an autoclave chamber. Finally it is subjected to HIP (Hot Isostatic Pressing).



Pressure distribution curve

8. EXPLOSIVE COMPACTING:

In this method, the pressure generated by an explosive is used to compact the metal powder.

Metal powder is placed in water proof bags which are immersed in water container cylinder of high wall thickness.

Due to sudden detonation of the charge at the end of the cylinder, the pressure of the cylinder increase.

This pressure is used to press the metal powder to form green compact.

9. FIBRE METAL PROCESS

In this process the metal fibres are mixed with a liquid slurry and poured over a porous bottom.

The liquid is drained off leaving the green mat of fibre .

The mat in which fibers are randomly distributed is pressed and sintered.

The products are mainly used for filters, battery plates.

10.HOT PRESSING

This involves applying pressure and temperature simultaneously, so that the compacting and sintering of the powder takes place at the same time in a die.

Its application is limited and can be used for compacting. Fe and Brass powders at much lower pressure than conventional pressing and sintering operations.

SPARK SINTERING

It is the process of simultaneous pressing and sintering of metal powders by high energy electric spark. The spark is discharged from capacitor bank and removes the surface contaminations from powder particles. This causes the particles to form solid cohesive mass.

Immediately after spark the current is continued for 10 seconds, with high temperature which further strengthens the bond. This method is best suitable for sintering Al,Cu,Fe and SS.

SINTERING (Very important)

Sintering is the operation of heating the green compact at high temperature in a controlled atmosphere containing Ar or H₂ gas (reducing atmosphere which protects oxidation of metal powders).

The sintering increases the bond between the particles and therefore strengthens a powder metal compact.

Sintering temperature & time of sintering depends on the type of powders and the strength required in a final product. Furnace temperature also controls sintering

Sintering temperature is usually 0.6 to 0.8 times the melting point of powders.

But in case of mixed powders of widely different melting temperature, the sintering temperature will usually above the melting point of minor constituent. (Example: Cobalt in cemented carbide & Cu in W).

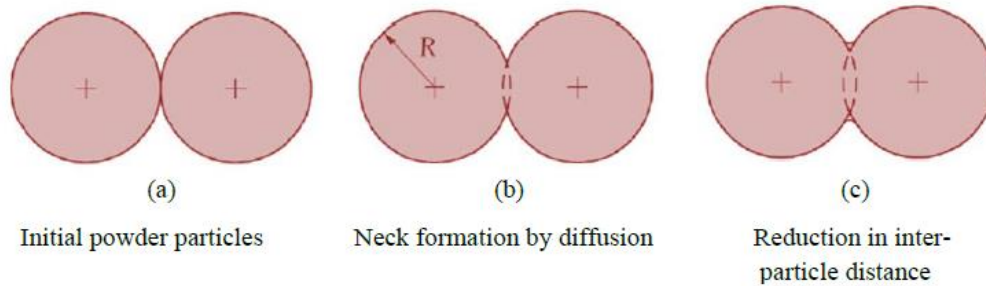
<i>Type of powders</i>	<i>Sintering temperature °C</i>	<i>Sintering Time</i>
Al & its alloys	370-520	24 hrs.
Cu, Brass & Bronze	700-900	30 min.
Iron	1025-1200	30 min.
Stainless Steel	1180	20-40 min
Tungsten Carbide	1480	20-40 min

MECHANISMS OF SINTERING

1. Diffusion(solid state sintering)
2. Vapour- phase transport
3. Liquid phase sintering

1. Diffusion(solid state sintering)

In this mass transfer occur at high temperature. At high temperature 2 adjacent particles begin to form a bond by diffusion and mass transport between 2 particles. As a result strength, density and thermal conductivity of the compact increases. But as the compact becomes denser, it shrinks in size.



Schematic illustration of sintering of compact preform using solid-state diffusion between powder particles

2. Vapour- phase transport

As the material is heated to temperature closer to melting point, some atoms may evaporate to vapour phase.

At the interphase of particles, the condition would be favorable for the vapor to condense and solidify.

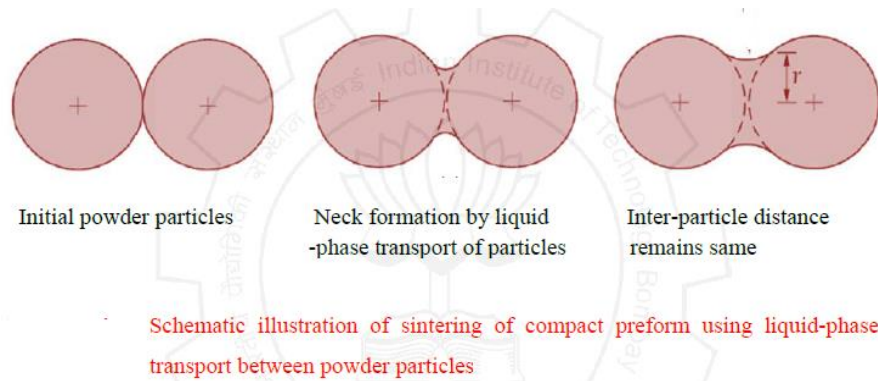
Thus the particles are joined by the formation of bond and particle shrink. This mechanism is not significant in most of the materials.

3. Liquid Phase sintering

The green compact may contain powders of different metals having different melting temperature. In such cases the metal with lower melting point may melt and surround the metal particles which are not melted.

This occurs due to surface tension of liquid metal. This mechanism is known as liquid phase sintering.

Stronger and denser products are obtained through this mechanism.

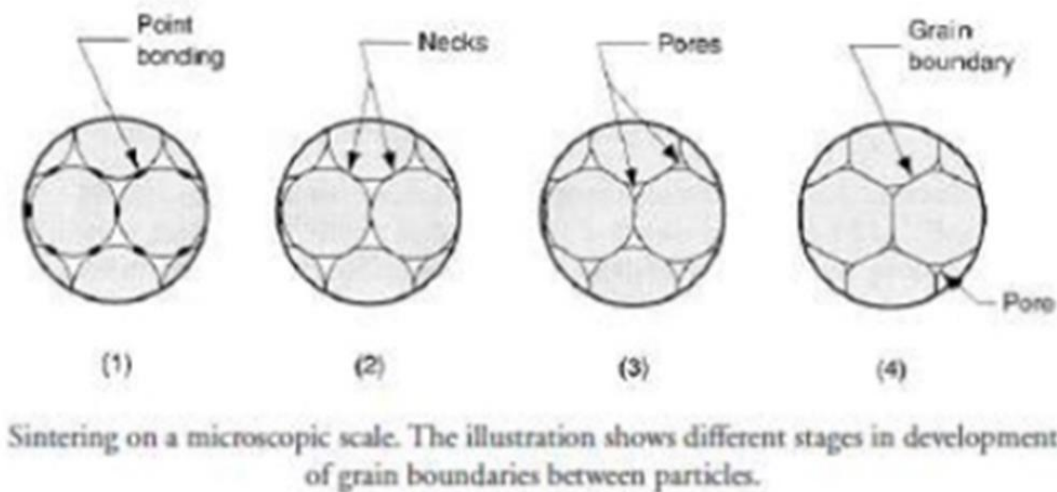


STAGES IN SINTERING (Very Important)

1. Preheating - Burning of additives
2. Temperature is raised to sintering temperature
3. Temperature is lowered

PORE SHRINKING (Very Important)

During sintering stage of a powder metallurgy product, mass transport between particle happens. This creates necking and new grain boundaries are formed. The size of pores are reduced by the formation of new grain boundaries. This is known as pore shrinking.

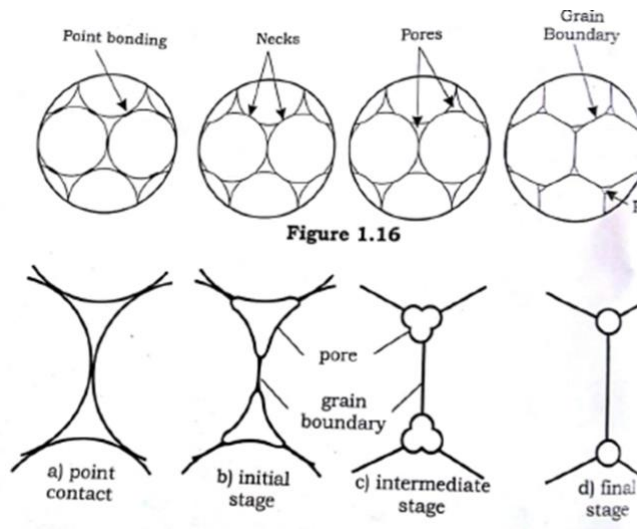


DRIVING FORCE FOR PORE SHRINKING

The primary driving force for pore shrinking is the reduction in surface energy of powder particles.

In the green compact condition total surface area of this particle is very high.

During sintering, by the application of heat the particles are bonded together and surface area is reduced. Hence the surface energy is also reduced. Due to this pore shrinkage happens.



SECONDARY OPERATIONS OR FINISHING OPERATIONS IN P/M

Secondary operations intended to provide dimensional tolerances, physical and better surface finish. They are

- | | |
|-----------------|-------------------|
| 1. Sizing | 5. Infiltration |
| 2. Coining | 6. Heat treatment |
| 3. Machining | 7. Plating |
| 4. Impregnation | |

Sizing: It is repressing the sintered component in the die to achieve the required accuracy.

Coining: It is repressing the sintered components in the die to increase density and to give additional strength.

Machining: Machining operation is carried out on sintered part to provide under cuts, holes, threads etc. which cannot be removed on the part in the powder metallurgy process. Cutting tools can be used for this purpose.

Impregnation: It is filling of oil, grease or other lubricants in a sintered component. Self-lubricating bearings, gears can be made by oil impregnation. This can be achieved by immersing the sintered product in a bath of hot oil.

Infiltration: It is filling of pores of sintered product with molten metal to improve physical properties.

Heat Treatment: The process of heating and cooling sintered parts at a desired rate to improve

- (i) Wear Resistance
- (ii) Grain Structure
- (iii) Strength

The following heat treatment processes are used to the parts made by powder metallurgy:

1. Stress relieving
2. Carburising
3. Nitriding
4. Induction Hardening

Plating: Plating is carried out in order to obtain corrosion resistance and better appearance

1. Import a pleasing appearance (Cr plating)
2. Protect from corrosion (Ni plating)
3. Improve electrical conductivity (Cu and Ag plating)

ADVANTAGES OF POWDER METALLURGY(V.Imp)

1. Wide range of properties such as density, porosity and particle size can be obtained for particular applications
2. Components can be produced with good surface finish and close tolerance.
3. High production rates.
4. Complex shapes can be produced

5. No wastage of raw material
6. Porous parts can be produced that could not be made any other way.
7. Impossible parts (cutting tool bits) can be produced.
8. Highly qualified or skilled labour is not required.
9. This process facilitates mixing of both metallic and non-metallic powders to give products of special characteristics.
10. There is usually no need for subsequent machining or finishing operations.

LIMITATIONS OF POWDER METALLURGY PROCESS(V .Imp)

1. The metal powders and the equipment used are very costly.
2. Storing of powders offer great difficulties because of possibility of fire and explosion hazards.
3. Parts manufactured by this process have poor ductility.
4. Sintering of low melting point powders like lead, zinc, tin etc., offer serious difficulties.
5. Strength and toughness of the parts produced are inferior to the forgings.
6. Difficult to produce parts with high uniform density.
7. Product size is limited.

APPLICATIONS OF POWDER METALLURGY(V.Imp)

1. Self-Lubricating Bearing and Filters:

Porous bronze bearings are made by mixing copper and tin powder in correct proportions, cold pressed to the desired shape and then sintered.

These bearings soak up considerable quantity of oil.

Hence during service, these bearings produce a constant supply of lubricant to the surface due to capillary action. These are used where lubricating is not possible.

2. **Porous filters** can be manufactured and are used to remove, undesirable materials from liquids and gases.
3. **Friction Materials:** These are made by powder metallurgy. Clutch liners and Brake bands are the example of friction materials.
4. **Gears and Pump Rotors:** Gears and pump rotor for automobile oil pumps are manufactured by powder metallurgy. Iron powder is mixed with graphite, compacted under a pressure of 40 kg/cm² and sintered in an electric furnace with an atmosphere and hydrocarbon gas. These are impregnated with oil.

5. **Refractory Materials:** Metals with high melting points are termed as refractory metals. These basically include four metals tungsten, molybdenum, tantalum and niobium. Refractory metals as well as their alloys are manufactured by powder metallurgy.

The applications are not limited to lamp filaments and heating elements, they also include space technology and the heavy metal used in radioactive shielding.

6. **Electrical Contacts and Electrodes:** Electrical contacts and resistance welding electrodes are made by powder metallurgy.
7. **Magnet Materials:** Soft and permanent magnets are manufactured by this process. Soft magnets are made of iron, iron-silicon and iron-nickel alloys. These are used in D.C. motors, or generators as armatures and in measuring instruments.

Permanent magnets known as Alnico which is a mixture of nickel, aluminium, cobalt, copper and iron are manufactured by this technique.

8. **Cemented Carbides:** These are very important products of powder metallurgy and find wide applications as cutting tools, wire drawing dies and deep drawing dies. These consist of carbides of tungsten, tantalum, titanium and molybdenum.
9. **Diamond Impregnated Tools**

These are made from a mixture of iron powder and diamond dust. Diamond dust acts as a cutting medium and iron powder acts as the bond. These tools are used for cutting porcelain and glass. These bits are welded or brazed to a steel shank.

INTRODUCTION TO METAL CUTTING

In Engineering industry, components are made of metals in different shapes, sizes and dimensions. Metals are shaped to the required forms by various processes. These processes can be generally divided into 2 groups.

1. Non cutting shaping process.
2. Cutting shaping process.

In non cutting shaping process the metal is shaped under the action of heat, pressure or both. Here there is no chip formation. This group includes operations like forging, drawing, spinning, drawing, extrusion etc.

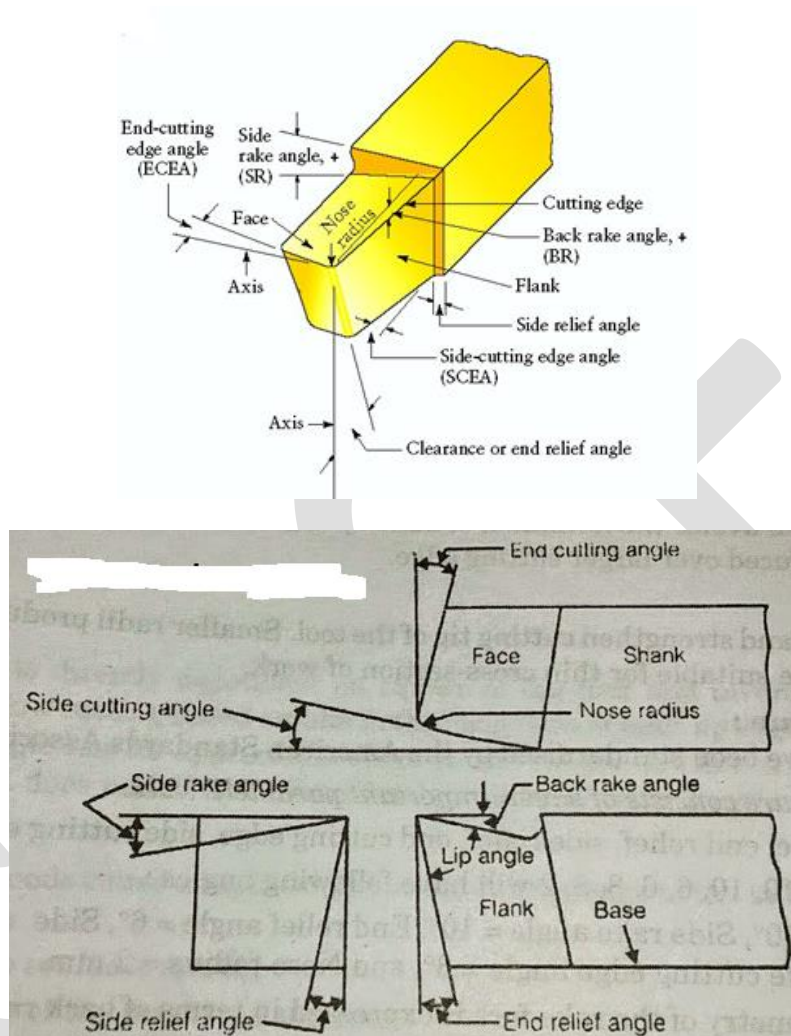
In cutting shaping process the required shape of metal is obtained by removing the unwanted material from the work piece in the form of chips. This group includes operations like turning, boring, milling, drilling, shaping, broaching etc. These operations are known as machining or metal cutting operations.

Machining Definition

Machining is a manufacturing process in which jobs are produced to the desired shape, dimensions and surface finish by gradually removing the excess material from the preformed blank in the form of chips with the help of cutting tools moved past the work surface.

TOOL NOMENCLATURE

Nomenclature means systematic naming of the various parts and angles of a cutting tool.



Part	Description
Shank	It is the body of the tool which is ungrounded.
Face	It is the surface over which the chip slides.
Base	It is the bottom surface of the shank.
Flank	It is the surface of the tool facing the work piece. There are two flanks namely end flank and side flank.
Cutting edge	It is the junction of the face and the flanks. There are two cutting edges namely side cutting edge and end cutting edge.
Nose	It is the junction of side and end cutting edges.

Important Angles in a single point cutting tool

1. Back rake angle

The back rake angle is the angle between the face of the tool and a line parallel to the base of the shank in a plane parallel to the side cutting edge.

It is the slope given to the face of the tool. Slope is given from nose along the length of the tool.

2. Side rake angle

It is the angle by which the face of the tool is inclined sideways.

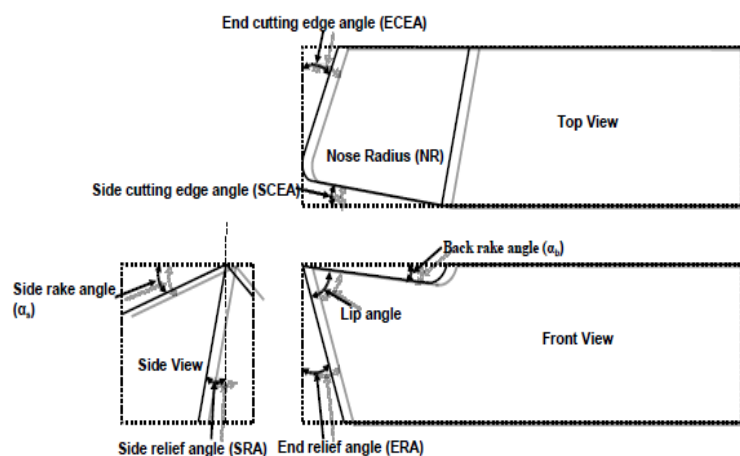
It is also the slope given to the face of the tool.

Slope is given from nose along the width of the tool

The side rake angle and the back rake angle combine to form the effective rake angle. This is also called true rake angle or resultant rake angle of the tool.

Positive Rake angle

Geometry of positive rake single point cutting tool



Positive rake or increased rake angle reduces compression, the forces, and the friction, yielding a thinner, less deformed and cooler chip.

But increased rake angle reduces the strength of the tool section, and heat conduction capacity.

Positive rake angles is recommended

- Machining low strength material
- Low power machine
- Long shaft of small diameter
- Low cutting speed

Negative Rake angle

To provide greater strength at the cutting edge and better heatconductivity, zero or negative rake angles are employed on carbide, ceramic, polycrystalline diamond, and polycrystalline cubic boron nitride cutting tools.

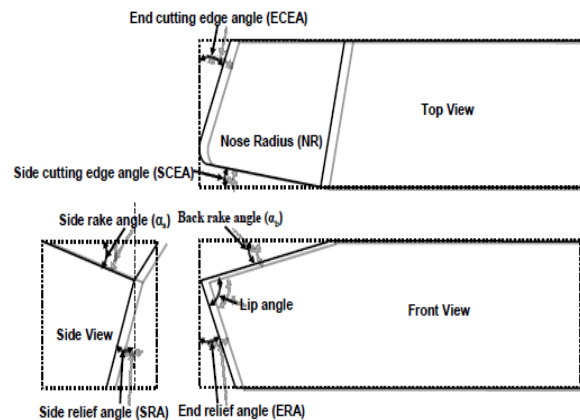
Negative rakes increases tool forces but this is necessary to provide added support to the cutting edge. Negative rake

- Increase edge strength
- Increases life of the tool
- Increases the cutting force
- High cutting speeds
- Requires ample power

Negative rake angles is recommended

- Machining high strength alloy
- Heavy impact loads
- High speed cutting
- With rigid set- up

Geometry of negative rake single point cutting tool



Zero rakeangle

- To simplify design and manufacturing of the form tools.
- Increases tool strength
- Avoids digging of the tool into the workpiece
- Brass is turned with zero rake angle

Relief angles or clearance angle

Relief angles are provided to minimize physical interference or rubbing contact with machined surface and the cutting tool.

1. Side relief angle
2. End relief angle

Small relief angles are essential when machining hard and strong materials and they should be increased for the weaker and softer materials.

3. Side relief angle:

It is the angle between side flank surface and a line perpendicular to the shank of the tool.

The Side relief angle prevents the side flank of the tool from rubbing against the work when longitudinal feed is given. Larger feed will require greater side relief angle.

4. End relief angle

It is the angle between end flank surface and a line perpendicular to the shank of the tool measured in a plane perpendicular to the base of the tool.

The End relief angle prevents the end flank of the tool from rubbing against the work.

Cutting edge angle

These are provided to form the cutting edge.

2 cutting edge angles are their

1. Side cutting edge angle

2. End cutting edge angle

5. **Side cutting edge angle** : It is the angle between the side cutting edge and axis of the tool

6. **End cutting edge angle**: It is the angle between the end cutting edge and perpendicular line to the axis of the tool

7. **Lip angle or cutting angle**: It is the angle between the face and end surface of the tool.

8. **Nose radius** It is the radius of the curved surface joins the side cutting edge & end cutting edge

ATTRIBUTES OF EACH TOOL NOMENCLATURE

1. Rake angle

The rake angle has the following function:

- It allows the chip to flow in convenient direction.
- It reduces the cutting force required to shear the metal and consequently helps to increase the tool life
- To reduce the power consumption for cutting.
- It provides sharpness to the cutting edge.
- It improves the surface finish
- Normally rake angle provided in a single point cutting tool is 5 to 10 deg.

2. Clearance angle

- Reduce tool wear
- Increase tool life.
- Must be positive (3deg – 15deg.)

3. Cutting Edge angle

Side cutting edge angle The following are the advantages of increasing this angle, ϕ

- It increases tool life as, for the same depth of cut; the cutting force is distributed on a wider surface ϕ
- It diminishes the chip thickness for the same amount of feed and permits greater cutting speed. ϕ It dissipates heat quickly for having wider cutting edge.
- The side cutting edge angle of the tool has practically no effect on the value of cutting force or power consumed for a given depth of cut and feed.
- ϕ Large side cutting edge angles are likely to cause the tool to chatter.
- A large end cutting edge angle unnecessarily weakens the tool. It varies from 8 to 15 degrees.

4. Nose radius (Functions of Nose Radius)

- Greater nose radius clears up the feed marks caused by the previous shearing action and provides better surface finish.
- It provides strengthening of the tool nose and better surface finish..
- If nose radius increased cutting force and cutting power increased.
- All finish turning tool have greater nose radius than rough turning tools

Tool signature

- It is the system of designating the principal angles of a single point cutting tool.
- The signature is the sequence of numbers listing the various angles, in degrees, and the size of the nose radius.
- There are several systems available like American standard system (ASA), orthogonal rake system (ORS), Normal rake system (NRS), and Maximum rake system (MRS).
- The system most commonly used is American Standard Association.

(ASA), which is:

- Back rake angle, Side rake angle, End relief angle, Side relief angle, End cutting Edge angle, Side cutting Edge angle and Nose radius.

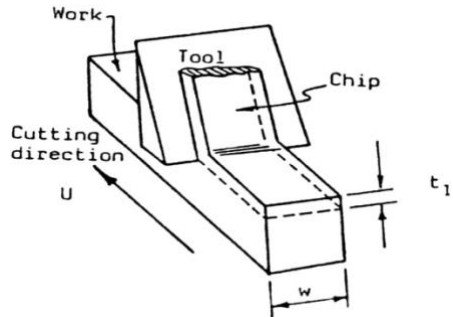
For example a tool may designated in the following sequence:

8-14-6-6-6-15-1

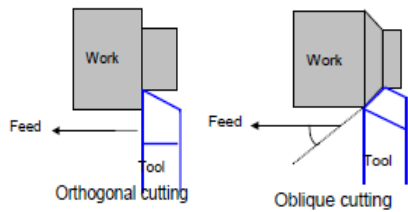
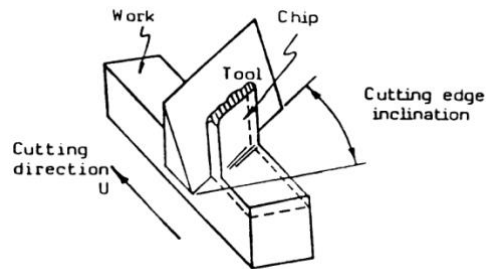
1. Back rake angle is 8
2. Side rake angle is 14
3. End relief angle is 6
4. Side relief angle is 6
5. End cutting Edge angle is 6
6. Side cutting Edge angle is 15
7. Nose radius is 1 mm .

METHODS OF METAL CUTTING

Orthogonal Cutting



Oblique Cutting



Difference between Orthogonal(2 D) Cutting & Oblique (3D) Cutting

Orthogonal metal cutting	Oblique metal cutting
Cutting edge of the tool is perpendicular to the direction of tool travel.	The cutting edge is inclined at an angle less than 90° to the direction of tool travel.
The direction of chip flow is perpendicular to the cutting edge.	The chip flows on the tool face making an angle.
The chip coils in a tight flat spiral	The chip flows side ways in a long curl.
For same feed and depth of cut the force which shears the metal acts on a smaller areas. So the life of the tool is less.	The cutting force acts on larger area and so tool life is more.
Produces sharp corners.	Produces a chamfer at the end of the cut
Smaller length of cutting edge is in contact with the work.	For the same depth of cut greater length of cutting edge is in contact with the work.

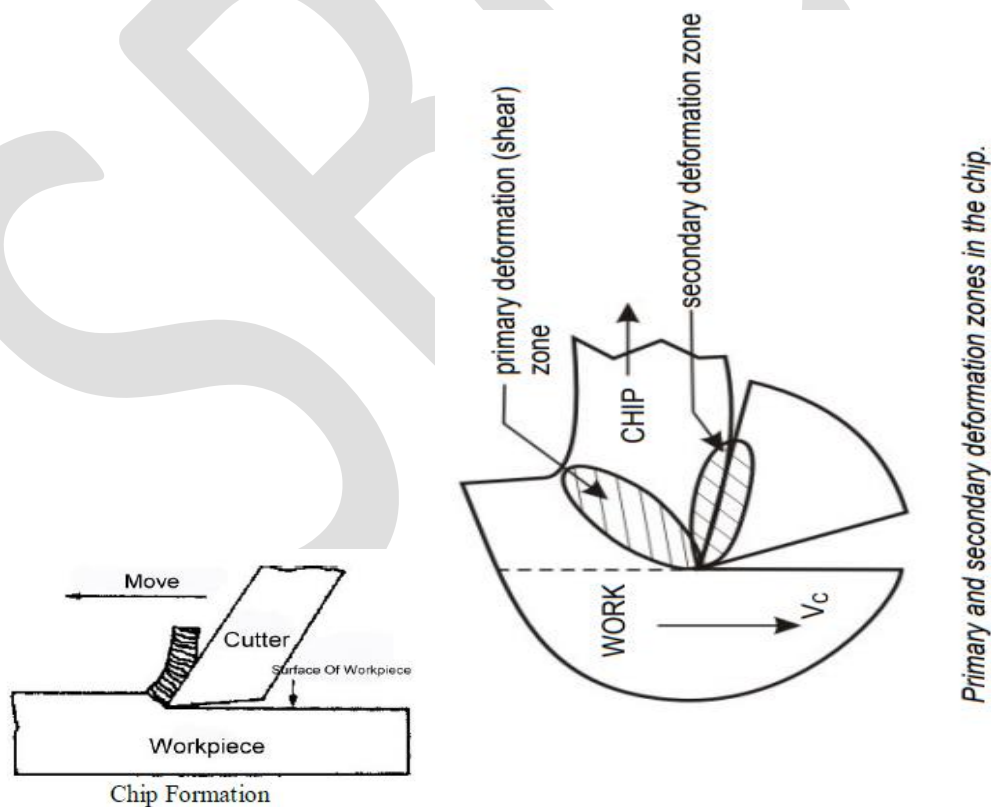
MECHANISM OF CHIP FORMATION

When the tool advances into the work piece, the metal in front of the tool is severely stressed. The cutting tool produces internal shearing action in the metal. The metal below the cutting edge yields and flows plastically in the form of chip. Compression of the metal under the tool takes place. When the ultimate stress of the metal is exceeded, separation of metal takes place. The plastic flow takes place in a localized area called as shear plane. The chip moves upward on the face of the tool.

The outward or shearing movement of each successive element is arrested by work hardening and the movement transferred to the next element.

The process is repetitive and a continuous chip is formed.

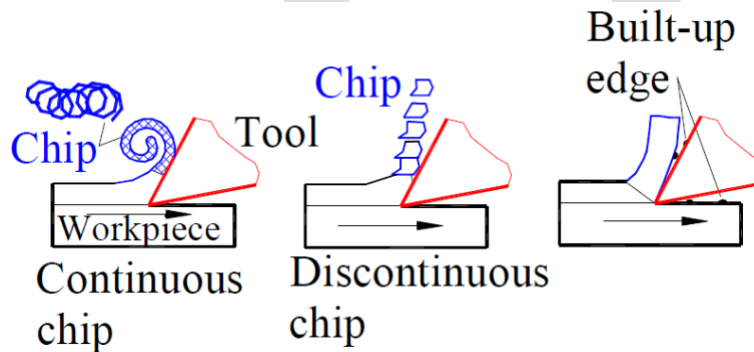
The plane along which the element shears, is called shear plane



TYPES OF CHIPS

There are three different types of chips. They are

1. Continuous chips,
2. Discontinuous chips
3. Chips with built up edge(BUE)



Types of chip depends on

- Work material (ductile, brittle)
- Cutting tool geometry (rake angle, cutting angle etc.)
- Cutting velocity and feed rate.
- Types of cutting fluid and method of application.

Continuous chips

- When the cutting tool moves towards the work piece there occurs a plastic deformation of the work piece and the metal is separated without any discontinuity and it moves like a ribbon.

- The conditions that favor the production of continuous chips is small chip thickness, high cutting speed, sharp cutting edge, large rake angle in cutting tool and fine feed, smooth tool face and efficient lubricating system.
- Such chips are produced while machining ductile materials like mild steel, copper and aluminum. Because of plastic deformation of ductile material long and continuous chips are produced.
- This is desirable because it produces good surface finish, low power consumption and longer tool life.
- These chips are difficult to handle and dispose off. Further the chips coil in a helix and curl around work and tool and may injure the operator when it is breaking. The tool face is in contact for a longer period resulting in more frictional heat. However this problem could be rectified by the use of chip breakers.

Discontinuous chips :

- These chips are produced when cutting more brittle materials like bronze, hard brass and gray cast iron.
- Since these chips break up into small segments the friction between chip and tool reduces resulting in better surface finish.
- These are convenient to handle and dispose off.
- Discontinuous chips are produced in ductile materials under the conditions such as large chip thickness, low cutting speed, small rake angle of tool etc.
- If these chips are produced from brittle materials, then the surface finish is fair, power consumption is low and tool life is reasonable however with ductile materials the surface finish is poor and tool wear is excessive .

Chips with built up edge(BUE) :

- This is nothing but a small built up edge sticking to the nose of the cutting tool. These built up edge occurs with continuous chips.

- When machining ductile materials due to conditions of high local temperature and extreme pressure the cutting zone and also high friction in the tool chip interface, there are possibilities of work material to weld to the cutting edge of tool and thus forming built up edges.
- This weld metal is extremely hard and brittle. This welding may affect the cutting action of tool.
- Successive layers are added to the build up edge. When this edge becomes large and unstable it is broken and part of it is carried up the face of the tool along with chip while remaining is left in the surface being machined. Thus contributing to the roughness of surface.
- Conditions favoring the formation of build up edge are low cutting speed, low rake angle, high feed and large depth of cut, ductile material, Cutting fluid absent or in adequate. This formation can be avoided by the use of coolants and taking light cuts at high speeds. This leads to the formation of crater on the surface of the tool.

Reduction or Elimination of BUE

Increase

- ↑ Cutting speed
- ↑ Rake angle
- ↑ Ambient work piece temperature.

Reduce

- ↓ Feed
- ↓ Depth of cut

Use

- Cutting fluid
- Change cutting tool material (as cermets).

CHIP BREAKERS

During machining, long and continuous chip will affect machining. It will spoil tool, work and machine. It will also be difficult to remove metal and also dangerous. The chip should be broken into small pieces for easy removal, safety and to prevent damage to machine and work. The function of chip breakers is to reduce the radius of curvature of chips and thus break it.

Chip breakers are used to break the long continuous chip into small pieces .The chip breaker is provided on cutting tool.

Types of chip breakers

The chip breaker classified into groove type and obstruction type

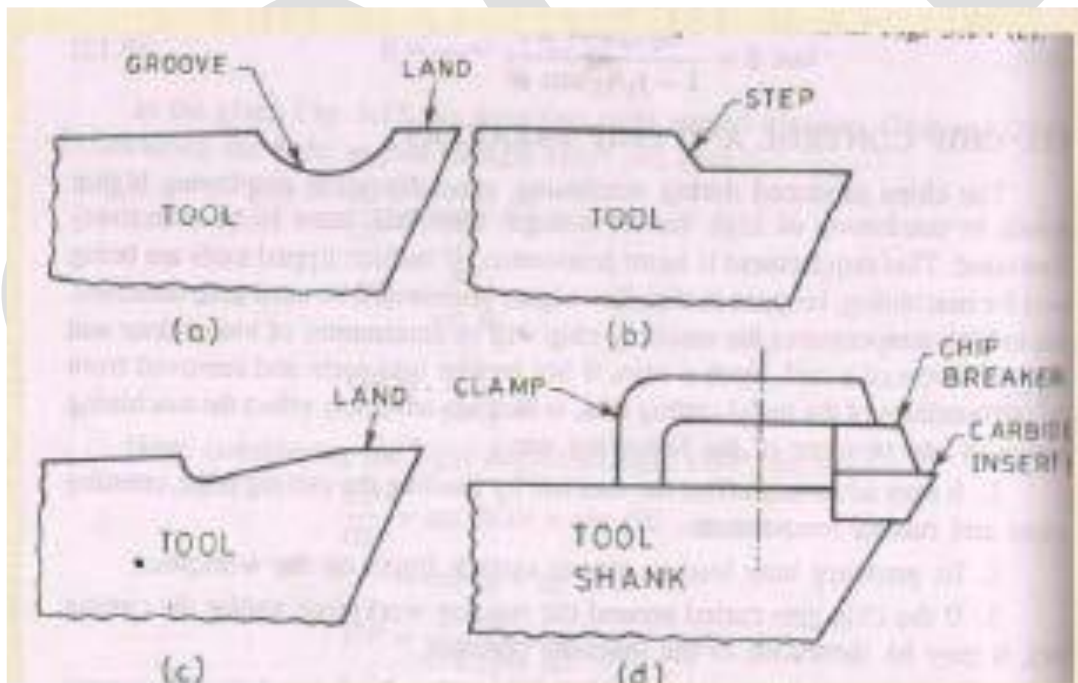
1.Groove type chip breaker: A small groove is provided behind the leading cutting edge of the tool insert on the rake face . The geometry of groove determines the radius of chip curvature.

2. Step type

In this a step is ground on the face of the tool along the cutting edge.

3.Clamp type

In this a thin carbide plate or clamp is brazed or screwed on the face of the tool.

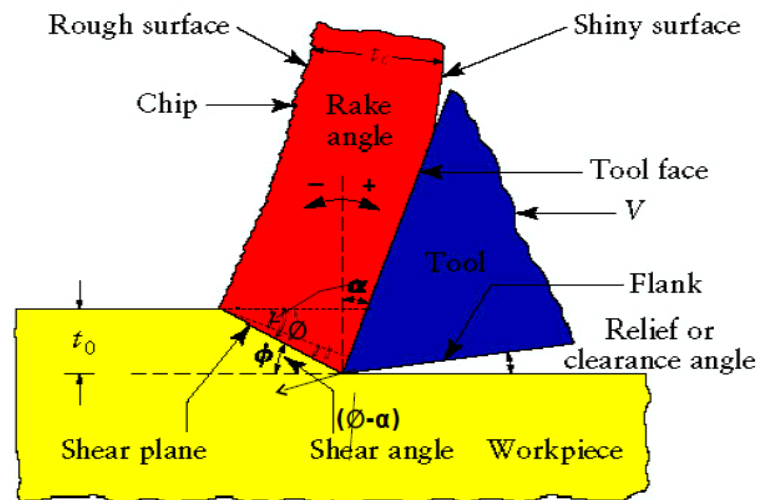


ANALYSIS OF CUTTING FORCES IN ORTHOGONAL METAL CUTTING

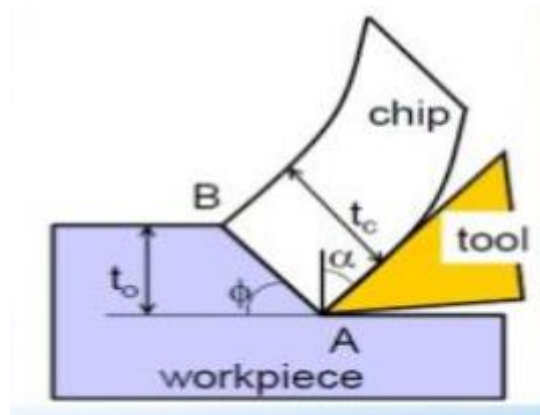
Assumptions in orthogonal metal cutting

1. No contact at the flank i.e. the tool is perfectly sharp.
2. No side flow of chips i.e. width of the chips remains constant.
3. Uniform cutting velocity.
4. A continuous chip is produced with no built up edge.
5. The chip is considered to be held in equilibrium by the action of the two equal and opposite resultant forces R and R_1 and assume that the resultant is collinear.

Metal cutting Terminologies



Chip thickness ratio (r) or cutting ratio.



The outward flow of the metal causes the chip to be thicker after the separation from the parent metal. That is the chip produced is thicker than the depth of cut.

Let t_o = chip thickness before cutting (uncut chip thickness)

t_c = chip thickness after cutting (cut chip thickness)

Then **Chip thickness ratio (r) = (t_o / t_c)**

Chip thickness ratio is always less than unity. If the ratio “ r ” is large the cutting action is good.

Chip reduction co efficient, **$K = (1 / r)$**

When metal is cut there is no change in volume of the metal. In orthogonal cutting the width of the chip equals the width of the cut. Considering specific gravity of the metal as constant, the volume of the chip produced will be equal to the volume of the metal cut. Width of both being equal, the product of the chip thickness and its length will, therefore be equal to the product of the thickness of the metal cut and length of metal cut. If L_1 and L_2 are lengths of the metal cut and chip respectively.

$$t \times L_1 = t_c \times L_2$$

$$\frac{t}{t_c} = \frac{L_2}{L_1} \quad \text{But} \quad \frac{t}{t_c} = r$$

$$r = \frac{t}{t_c} = \frac{L_2}{L_1}$$

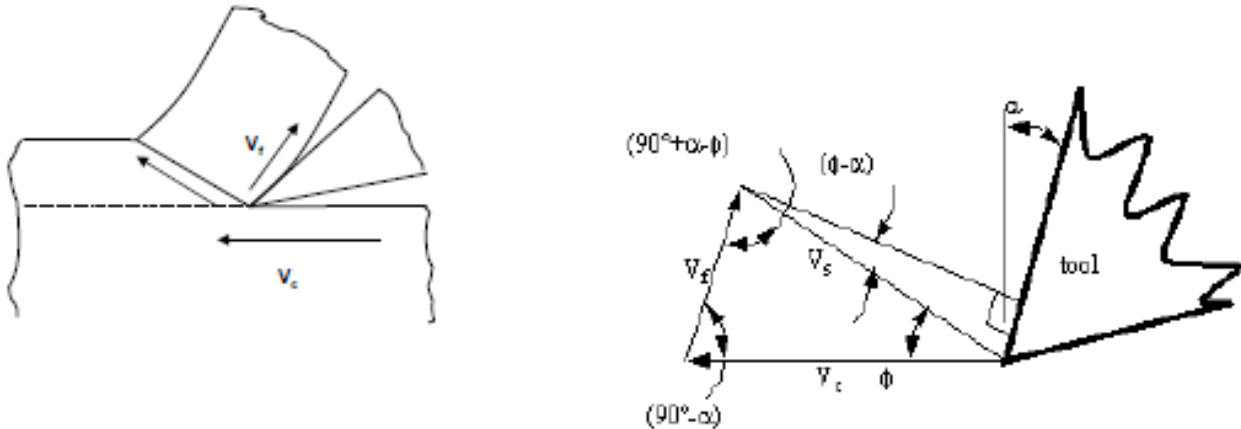
$$k = \frac{1}{r} = \frac{t}{t_c} = \frac{L_2}{L_1}$$

Derive the relation for shear angle in orthogonal metal cutting operation

SBCE

VELOCITY RELATIONSHIP IN ORTHOGONAL METAL CUTTING

The relationship of different velocities in orthogonal metal cutting is shown in figure. Let the velocities are shown in figure



V_c = Velocity of tool relative to work or the cutting velocity

V_f = Velocity of chip flow relative to tool

V_s = Velocity of displacement of the chip along the shear plane relative to work ,or the velocity of shear.

Of the above 3 velocities cutting velocity V_c is always known . The other 2 can be computed with the help of the following relations.

Analytically,

$$\frac{v_c}{\sin(90 - (\phi - \alpha))} = \frac{v_f}{\sin \phi} = \frac{v_s}{\sin(90 - \alpha)}$$

$$\frac{v_c}{\cos(\phi - \alpha)} = \frac{v_f}{\sin \phi} = \frac{v_s}{\cos \alpha}$$

where,

V_c = cutting velocity (ft./min.) - as set or measured on

V_s = shearing velocity

V_f = frictional velocity

$$v_f = \frac{v_c \sin \phi}{\cos(\phi - \alpha)} \quad \left[r = \frac{\sin \phi}{\cos(\phi - \alpha)} \right]$$

$$v_f = v_c \times r$$

$$v_s = \frac{v_c \cos \alpha}{\cos(\phi - \alpha)}$$

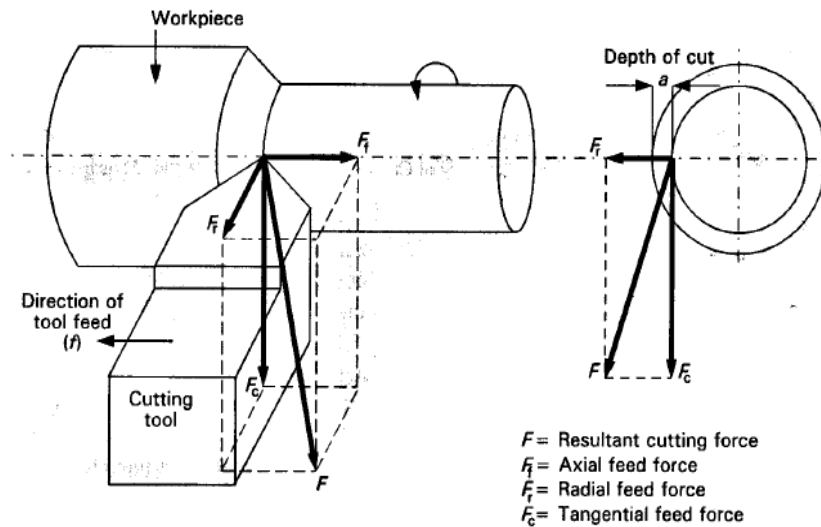
Volume of material per unit time = Volume of material flowing up the chip

$$\Rightarrow v_c \times t_0 \times w = v_f \times t_c \times w$$

$$\Rightarrow v_f = v_c \times r \quad \text{As, } r = \frac{t_0}{t_c}$$

CUTTING FORCES

The force system in general case of conventional turning process



The largest magnitude is the vertical force F_c which in turning is larger than feed force F_f , and F_f is larger than radial force F_r .

For orthogonal cutting system F_r is made zero by placing the face of cutting tool at 90 degree to the line of action of the tool.

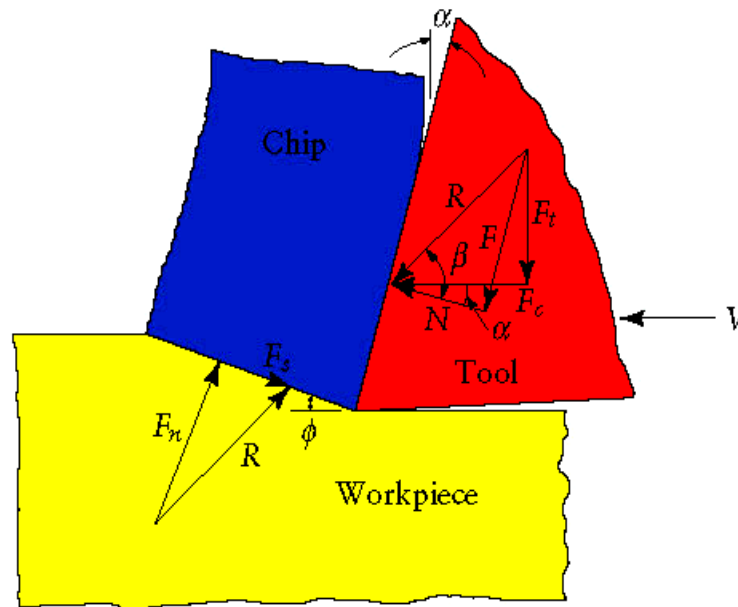
$$\begin{aligned}
 \bar{R} &= \bar{F}_x + \bar{F}_y + \bar{F}_z \\
 &= \bar{F}_c + \bar{F}_f + \bar{F}_r
 \end{aligned}$$

↑ ↑ ↑
 Tangential Feed Radial
 Force (Cutting Force Force
 In orthogonal cutting

$$\bar{R} = \bar{F}_c + \bar{F}_t$$

↑
 Thrust Force

Forces acting on Chip in two-dimensional cutting



F_s = Shear Force, which acts along the shear plane, is the resistance to shear of the metal in forming the chip.

F_n = Force acting normal to the shear plane, is the backing up force on the chip provided by the workpiece. (Compressive force)

F = Frictional resistance of the tool acting against the motion of the chip as it moves upward along the tool.

N = Normal to the chip force, is provided by the tool.

F_c = Cutting force

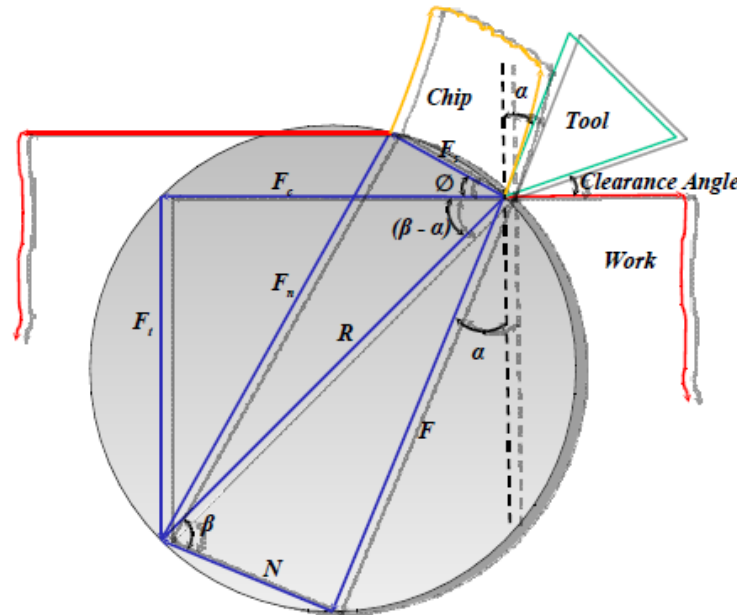
F_t = Feed force or thrust force

Merchant's Circle Diagram

The following is a circle diagram. Known as Merchant's circle diagram, which is convenient to determine the relation between the various forces and angles. In the diagram two force triangles have been combined and R and $R/$ together have been replaced by R . the force R can be resolved into two components F_c and F_t . F_c and F_t can be determined by force dynamometers.

$$R = F_c + F_t$$

The rake angle (α) can be measured from the tool, and force F and N can then be determined. The shear angle (ϕ) can be obtained from its relation with chip reduction coefficient. Now F_s & F_n can also be determined.



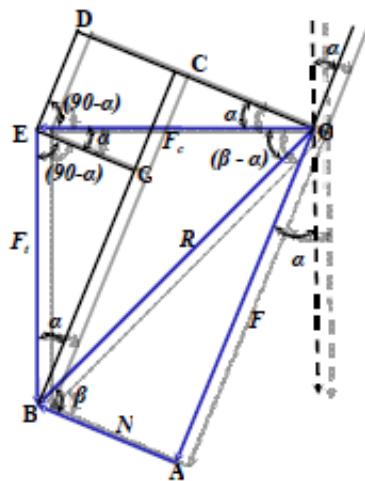
The procedure to construct a merchants circle diagram

- Set up x-y axis labeled with forces, and the origin in the centre of the page. The cutting force (F_c) is drawn horizontally, and the tangential force (F_t) is drawn vertically. (Draw in the resultant (R) of F_c and F_t).
- Locate the centre of R , and draw a circle that encloses vector R . If done correctly, the heads and tails of all 3 vectors will lie on this circle.
- Draw in the cutting tool in the upper right hand quadrant, taking care to draw the correct rake angle (α) from the vertical axis.
- Extend the line that is the cutting face of the tool (at the same rake angle) through the circle. This now gives the friction vector (F).
- A line can now be drawn from the head of the friction vector, to the head of the resultant vector (R). This gives the normal vector (N). Also add a friction angle (β) between vectors R and N . Therefore, mathematically, $R = F_c + F_t = F + N$.
- Draw a feed thickness line parallel to the horizontal axis. Next draw a chip thickness line parallel to the tool cutting face.

- Draw a vector from the origin (tool point) towards the intersection of the two chip lines, stopping at the circle. The result will be a shear force vector (F_s). Also measure the shear force angle between F_s and F_c .
- Finally add the shear force normal (F_n) from the head of F_s to the head of R .
- Use a scale and protractor to measure off all distances (forces) and angles

Relationship of various forces acting on the chip with the horizontal and vertical cutting force from Merchant circle diagram

Frictional Force System

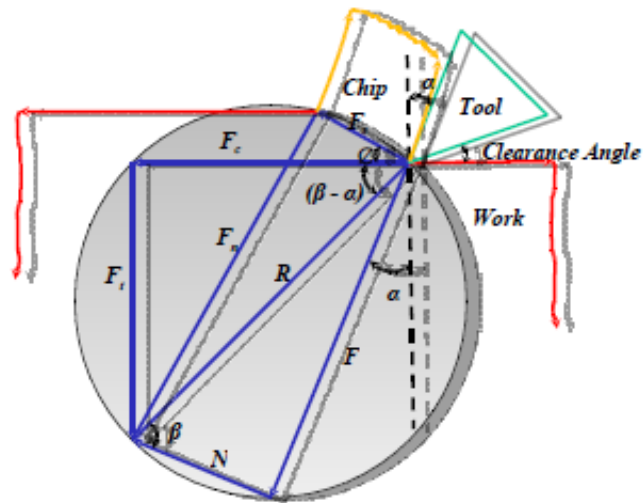


$$F = OA = CB = CG + GB = ED + GB$$

$$\Rightarrow F = F_c \sin \alpha + F_i \cos \alpha$$

$$N = AB = OD - CD = OD - GE$$

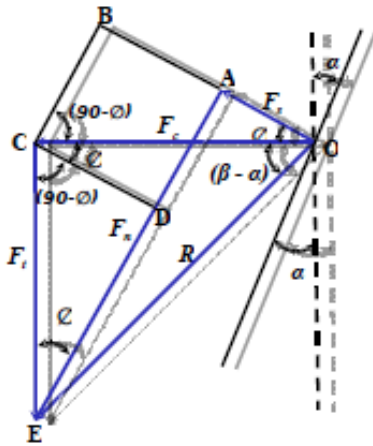
$$\Rightarrow N = F_c \cos \alpha - F_i \sin \alpha$$



The coefficient of friction

$$\mu = \tan \beta = \frac{F}{N}$$

Where β = Friction angle

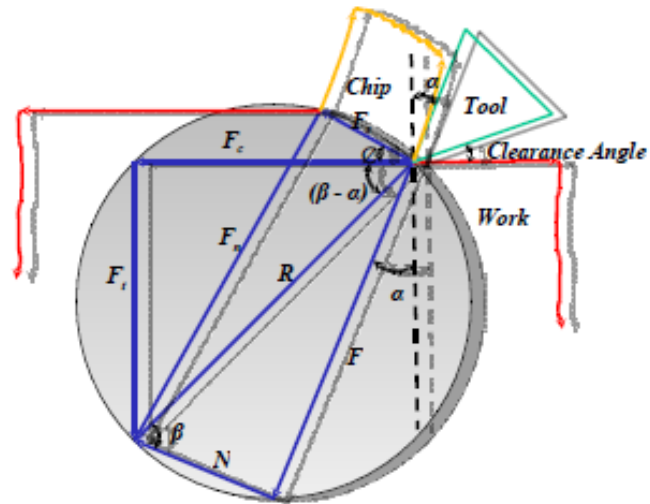
Shear Force System

$$F_s = OA = OB - AB = OB - CD$$

$$\Rightarrow F_s = F_c \cos \phi - F_t \sin \phi$$

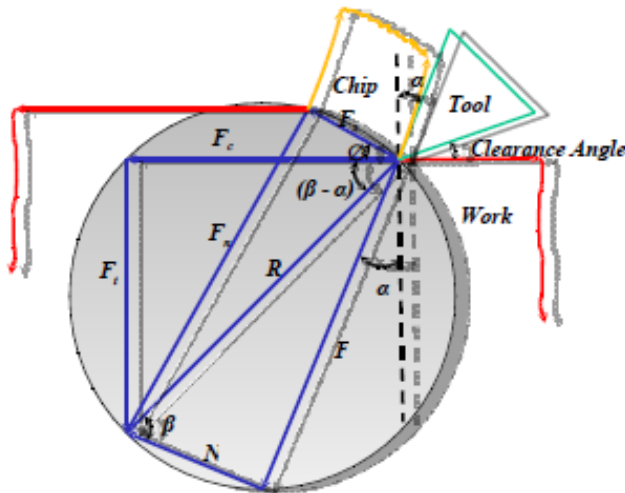
$$F_N = AE = AD + DE = BC + DE$$

$$\Rightarrow F_N = F_c \sin \phi + F_t \cos \phi$$



Also:

$$F_N = F_s \tan(\phi + \beta - \alpha)$$



$$F = F_c \sin \alpha + F_t \cos \alpha$$

$$N = F_c \cos \alpha - F_t \sin \alpha$$

$$F_s = F_c \cos \phi - F_t \sin \phi$$

$$F_N = F_c \sin \phi + F_t \cos \phi$$

$$F_N = F_s \tan(\phi + \beta - \alpha)$$

Theory of Ernst and Merchant (1944)

Ernest and Merchant gave the relation

$$\phi = \frac{\pi}{4} - \frac{1}{2}(\beta - \alpha)$$



M. Eugene Merchant

Assumptions of the theory:

- Tool edge is sharp.
- The work material undergoes deformation across a thin shear plane.
- There is uniform distribution of normal and shear stress on the shear plane.
- The work material is rigid and perfectly plastic.
- The shear angle ϕ adjusts itself to give minimum work.
- The friction angle β remains constant and is independent of ϕ .
- The chip width remains constant.

Theory of Ernst and Merchant (1944)

$$F_s = R \cos(\phi + \beta - \alpha)$$

$$R = F_c \sec(\beta - \alpha)$$

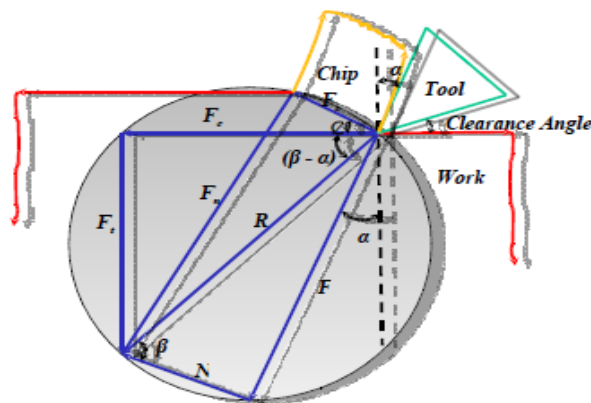
$$\Rightarrow F_s = F_c \sec(\beta - \alpha) \cos(\phi + \beta - \alpha)$$

$$\tau_s = \frac{F_s}{A_s}$$

$$\text{where, } A_s = \frac{wt_0}{\sin \phi}$$

$$\Rightarrow \tau_s = \frac{F_c \sec(\beta - \alpha) \cos(\phi + \beta - \alpha)}{\frac{wt_0}{\sin \phi}}$$

$$\Rightarrow \tau_s = \frac{F_c \sec(\beta - \alpha) \cos(\phi + \beta - \alpha) \sin \phi}{wt_0}$$



They have assumed that ϕ adjusts itself to give minimum work. And for a given set of cutting condition, to, w and α are all constants. They also assumed that β is independent of ϕ .

Theory of Ernst and Merchant (1944)

We can either maximize τ_s or minimize F_c

Therefore in the above equation the term $\cos(\phi + \beta - \alpha) \sin \phi$ contains only one variable ϕ .

let $y = \cos(\phi + \beta - \alpha) \sin \phi$

$$\frac{dy}{d\phi} = -\sin(\phi + \beta - \alpha) \sin \phi + \cos(\phi + \beta - \alpha) \cos \phi$$

for maximum value of y

$$\Rightarrow \frac{dy}{d\phi} = 0$$

$$\Rightarrow \sin(\phi + \beta - \alpha) \sin \phi = \cos(\phi + \beta - \alpha) \cos \phi$$

$$\Rightarrow \tan(\phi + \beta - \alpha) = \cot \phi$$

$$\Rightarrow \tan(\phi + \beta - \alpha) = \tan\left(\frac{\pi}{2} - \phi\right)$$

$$\Rightarrow \phi + \beta - \alpha = \frac{\pi}{2} - \phi$$

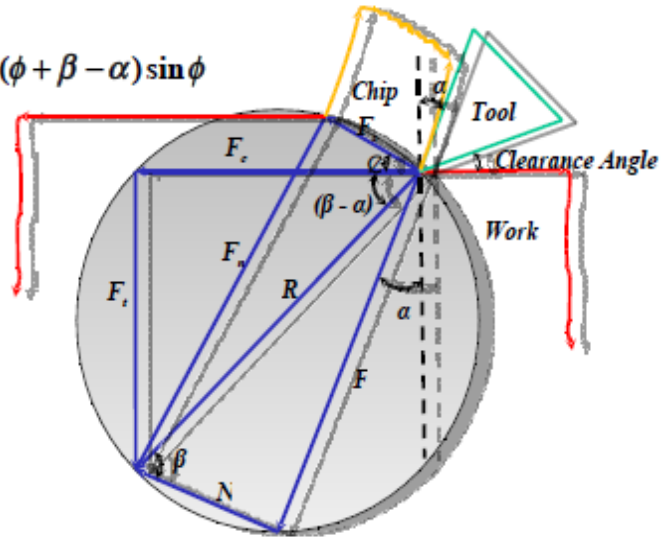
$$\Rightarrow \phi = \frac{\pi}{4} - \frac{1}{2}(\beta - \alpha)$$

Experimental verification revealed that the above equation is an over estimate.

Merchant later modified this equation and gave another equation $2\phi + \beta - \alpha = C$

Where C is the machining constant. Usually $C \leq \frac{\pi}{2}$ depends upon the work materials.

According to Merchant, C is a property of work material unaffected by cutting conditions, but grain size and micro structure have an affect on C .



CUTTING TOOL MATERIALS

Properties required for a Cutting tool material

1. Hardness:

The tool material must be harder than the work piece material.

Higher the hardness, easier it is for the tool to penetrate the work material

2. Hot hardness

Hot Hardness is the ability of the cutting tool must to maintain its Hardness and strength at elevated temperatures.

This property is more important when the tool is used at higher cutting speeds, for increased productivity

3. Toughness

In spite of the tool being tough, it should have enough toughness to withstand the impact loads that come in the start of the cut to force fluctuations due to imperfections in the work material

4. Wear Resistance

Wear resistance means the attainment of acceptable tool life before tools need to be replaced.

5. Low friction:

The coefficient of friction between the tool and chip should be low. This would lower wear rates and allow better chip flow.

6. Thermal characteristics:

Since a lot of heat is generated at the cutting zone, the tool material should have higher thermal conductivity to dissipate the heat in shortest possible time, otherwise the tool temperature would become high, reducing its life

The commonly used cutting tool materials are

1. Carbon steel
2. Medium alloy steel
3. High Speed Steel (1900)
4. Cemented Carbides(1926-30)
5. Ceramics (1950s)
6. Stellites
7. Abrasives
8. CBN(Cubic Boron Nitride) (1962)

9. Diamond

Carbon steel

- These are the oldest of the tool materials
- No sufficient hardness and wear resistance
- Low thermal resistance value
- Low hot hardness value (200- 250 °C)
- Limited to low cutting speed operation (12 m/min)
- Inexpensive, easily shaped, sharpened.
- Used in tools like-files, core drills ,short reamers etc
- Composition C 0.7 to 1%,Si 0.5%, Mn 0.5%

Medium alloy steel

- Similar to carbon steel with the addition of other elements like Cr,Mo,W
- Hot hardness value (250-300°C)
- Composition C up to 1.3 %, Si up to 0.4%, Mn 0.25-.75%
- Cr 0.4 to 0.8 % W 1 to 5 % Fe (balance)

High Speed Steel

- Mostly used in industries
- They can be used at higher cutting speed(2 to 3 times higher than for C Steel) . So it is called HSS.
- Hot hardness value (900°C)
- Good wear resistance
- High toughness
- Good for positive rake angle tools
- Widely used alloy steel
- Alloy of Tungsten, Chromium, Vanadium, Cobalt& Molybdenum
- They are of 3 types:
- Tungsten HSS (denoted by T), Molybdenum HSS (denoted by M). Cobalt HSS (Super HSS)

a) 18-4-1 HSS

W-18% Cr-4% V-1% with C-0.6 to 0.7

b) Cobalt HSS (Super HSS)

W-20% Cr-4% V-2% Co-12%

c) Molybdenum HSS

W-6% Cr-4% V-2% Mo-6%

Cemented Carbides

Composing of Carbon , Tungsten Carbide, Titanium Carbide.

Manufacturing

- Mixed (tungsten + pure carbon) at 1500 °C in the ratio of 94:6 by weight
- New compound(now tungsten carbide) mixed with cobalt to control the toughness of the tool
- Above homogeneous mixture pressed at 1000 to 4200 kg/cm²
- Then heated with the presence of H₂
- Cut into different shapes

Two Type

- Titanium –Tungsten type cemented carbide

Tungsten carbide 66 to 85 %

Titanium carbide 5 to 30 %

Cobalt 4 to 10 %

Wear resistance high

- Tungsten Type cemented carbide

Tungsten carbide 92% to 98%

Cobalt 2 to 8%

- Less brittle than Ti-W type (cemented carbide)
- Used for machine brittle material such as cast iron ,bronze
- Hot hardness value 1000 °C
- Used at cutting speed higher than 6 % that used with HSS
- Good thermal and wear resistance

Ceramics

Low heat conductivity(Used without coolant)

Quite brittle

Cannot be used for tools operating in interrupted cuts

High temp. Resistance (Up to 1200 °C)

Cutting speed more than 40 times of HSS

Manufacturing

Aluminium oxide powder is taken in mould and pressed at a pressure of 300 kg/cm² and then sintered at 2000 °C.

Tool tip is prepared and brazed or clamped to the shank of cutting tool

Stellites

Non ferrous cast Alloy

Cobalt 40-48%

Chromium 30 - 35%

Tungsten 10 -20 %

C-Content 1.8-2.5%

- High hardness
- Good wear resistance
- High temperature hardness (Up to 1000 °C)
- No Toughness
- Not widely used in metal cutting because it is too brittle.
- Used for cutting rubber, plastic

CBN (Cubic Boron Nitride)

- Hardest tool material available next to diamond
- This material consists of atoms of nitrogen and boron.
- High hardness, tensile strength, wear resistance
- Traded in the name of BOROZON
- A thin layer of (0.5 mm) CBN is applied to cemented carbide tools to improve machining performance

Diamond

- Hardest, incompressible
- Readily conducts heat, Low coefficient of friction
- Cutting speed 50 times greater than HSS
- Temp Resistance up to 1650 °C
- Used for cutting glass, plastics
- high wear resistance

Coated carbide tools

- Coated with layer of optimum thickness of Titanium carbide, Titanium carbonitride, and titanium nitride

- Layer coated is carried out by chemical vapour deposition process

Qn.Why ceramics are normally provided as inserts for tools, and not as entire tools?

Ceramics are brittle materials and cannot provide the structural strength required for a tool.

Qn.Explain the Influence of alloying elements in steel properties

Carbon: Carbon presence leads to the formation of carbides ,which contributes hardness and wear resistance.

Silicon, Manganese: Contributes improved hardness and strength of steel.

Tungsten or Molybdenum: Contributes red hardness

Vanadium: Contributes toughness, hardness.

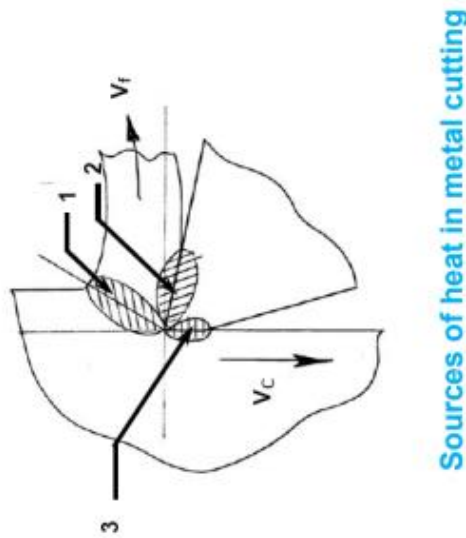
Chromium: Contributes corrosion resistance.

Cobalt: Provides improved toughness and tensile strength.

THERMAL ASPECTS OF MACHINING

According to the first law of thermodynamics, when work is transformed into heat, the quantity of heat produced is equivalent to the quantity of work. Heat generated, through conversion of mechanical energy. The three distinct sources of heat in metal cutting are given below:

- The shear zone, 1, where the primary plastic or shear deformation takes place
- The chip-tool interface, 2, where secondary plastic deformation due to friction between the heated chip and tool takes place.
- The work- tool interface, 3, at flanks where frictional rubbing occurs



For example, in a typical study of machining mild steel at 30 m/min at about 750 deg of cutting temperature at tool-chip interface, the distribution of total energy developed at the shear zone is as follows

- **Energy at chip – 60 percent**
- **Energy to workpiece – 30 percent**
- **Energy to tool - 10 percent**

The rate of energy consumption during orthogonal cutting is given by

$W_c = F_c V_c$ Where F_c = Cutting force , N

V_c = cutting speed , m/min

When a material is deformed elastically, the energy used is stored in the material as strain energy and no heat is generated. However, when a material is deformed plastically almost all the energy used is converted into heat.

In metal cutting , the material is subjected to extremely high strains and elastic deformation forms a very small proportion of the total deformation, hence all the energy is assumed to be converted into heat.

Thus $Q = F_c V_c / J$ where J is mechanical equivalent of heat

The cutting energy is converted into heat in two principal regions of plastic deformation The shear zone or primary deformation zone AB Secondary deformation zone BC.

EFFECTS OF THE HIGH CUTTING TEMPERATURE ON TOOL AND WORK

High cutting temperatures are detrimental to both the tool and the job. The major portion of the heat is taken away by the chips. But it does not matter because chips are thrown out. So attempts should be made such that the chips take away more and more amount of heat leaving small amount of heat to harm the tool and the job. The possible detrimental effects of the high cutting temperature on cutting are:

On tool

- Rapid tool wear , which reduces tool life
- Cutting edges plastically deform and tool may loose its hot hardness
- Thermal flaking and fracturing of cutting edges may take place due to thermal shock
- Built up edge formation

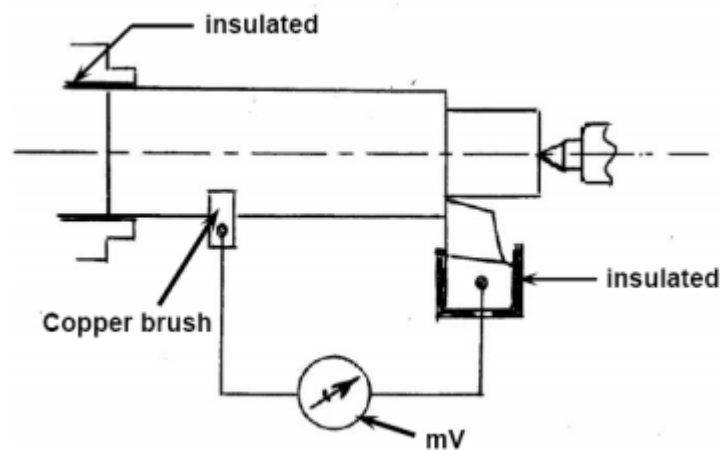
On work

- Dimension inaccuracy of work due to thermal distortion and expansion and contraction during and after machining
- Surface damage by oxidation, rapid corrosion, burning etc.
- Tensile residual stresses and microcracks at the surface and sub surfaces.

DETERMINATION OF CUTTING TEMPERATURE

1. Calorimetric method – quite simple and low cost but inaccurate and gives only grand average value
2. Decolourising agent – some paint or tape, which change in colour with variation of temperature, is pasted on the tool or job near the cutting point; the as such colour of the chip (steels) may also often indicate cutting temperature
3. Tool-work thermocouple – simple and inexpensive but gives only average or maximum value
4. Moving thermocouple technique
5. Embedded thermocouple technique
6. Photo-cell technique
7. Infra ray detection method

Tool work thermocouple technique



Tool work thermo couple set up for measurement of temperature.

In a thermocouple two dissimilar but electrically conductive metals are connected at two junctions. Whenever one of the junctions is heated, the difference in temperature at the hot and cold junctions produce a proportional current which is detected and measured by a milli-voltmeter.

In machining like turning, the tool and the job constitute the two dissimilar metals and the cutting zone functions as the hot junction as shown in Figure . Then the average cutting temperature is evaluated from the mV after thorough calibration for establishing the exact relation between mV and the cutting temperature.

TOOL WEAR

Mainly 4 types of wear

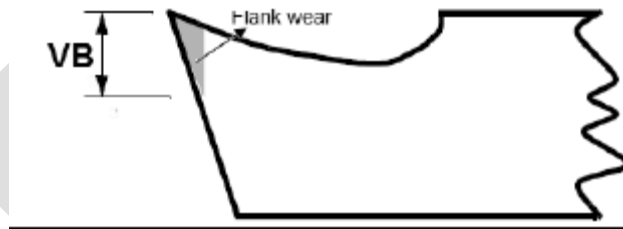
1. Flank Wear
2. Crater Wear
3. Chipping off of the cutting edge
4. Rounding of the cutting edge.

1. Flank Wear: Wear occurred on the flanks of the cutting tool.

Reasons

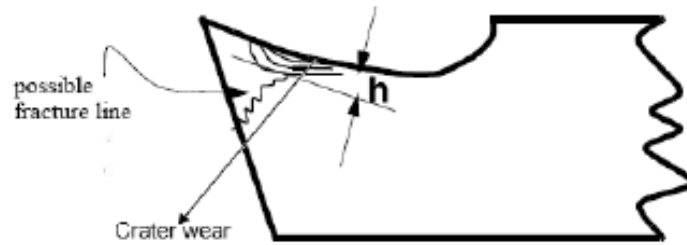
1. Abrasion by hard particles and inclusions in the work piece.
2. Shearing of the micro welds between tool and work material.
3. Abrasion by fragments of built-up-edge ploughing against the clearance face of the tool.
4. At low speed flank wear predominates.

If MRR increased flank wear increased.



2. Crater Wear

- 1) More common in ductile materials which produce continuous chip.
- 2) Crater wear occurs on the rake face of the cutting tool in the form of a crater.
- 3) At very high speed crater wear predominates
- 4) For crater wear temperature is main culprit and tool defuse into the chip material & tool temperature is maximum at some distance from the tool tip.



3. Chipping of cutting edge

- 1) Tool material is too brittle
- 2) Weak design of tool, such as high positive rake angle
- 3) As a result of crack that is already in the tool
- 4) Excessive static or shock loading of the tool

WEAR MECHANISMS

1. Abrasion wear:

- occurs when hard constituents of one surface moves to the other side of material
- Amount of wear depends on hardness of contacting surface.
- Constituents causing wear are harder than the matrix of the cutting tool.

2. Adhesion wear:

- occurs when two surfaces are brought into contact under normal loads and form welded junctions
- When shear load is applied, this junction breaks
- Depends on characteristics of tool and work material
- Occurs at very low cutting speeds

3. Diffusion wear:

- Atoms move from one lattice point to another
- Caused due to increase in temperature
- Occurs when localized temperature is increased sufficiently

4. Chemical or oxidation wear

- The fresh metal will quickly react with oxygen in air to form a fresh oxide layer,
- which will then be scraped off again by asperities in the following cycle.
- Such an "oxidation - scrape", or "chemical - mechanical" cycle repeats during the oxidation wear process, producing wear debris of finely powdered oxide

TOOL LIFE

Tool life is the time a tool will operate satisfactorily until it is dulled.

Tool failure is mainly happened as tool wear.

Taylor's Tool Life Equation

based on Flank Wear

Causes

- Sliding of the tool along the machined surface
- Temperature rise

$$VT^n = C$$

Where, V = cutting speed (m/min)

T = Time (min)

n = exponent depends on tool material

C = constant based on tool and work material and cutting condition.

Extended or Modified Taylor's equation

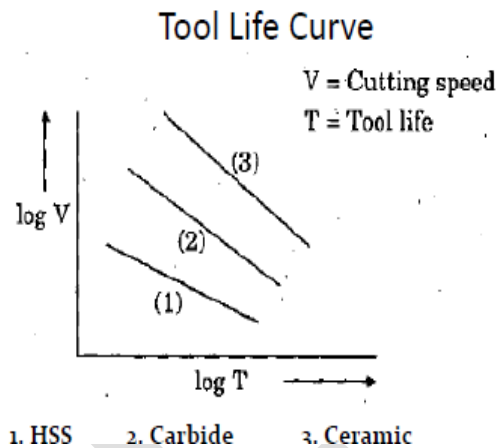
$$VT^n f^a d^b = C$$

Where: d = depth of cut

f = feed rate

Values of Exponent 'n'

$n = 0.08$ to 0.2 for HSS tool
 $= 0.1$ to 0.15 for Cast Alloys
 $= 0.2$ to 0.4 for carbide tool
 $= 0.5$ to 0.7 for ceramic tool



In the graph slope of the line represent the value of n

FACTORS AFFECTING TOOL LIFE

1. Cutting speed
2. Feed and depth of cut
3. Tool geometry
4. Tool material
5. Cutting fluid
6. Work material
7. Rigidity of work, tool, and machine

CUTTING FLUIDS

Cutting fluid may be defined as substance which is applied to a tool during a cutting operation to facilitate removal of chips and heat

FUNCTIONS OF CUTTING FLUIDS IN MACHINING

- 1) To lubricate chip, tool & work piece
- 2) To cool the cutting tool and work piece

- 3) To help carry away the chips from the cutting zone.
- 4) To lubricate some the moving parts of MT
- 5) To improve the surface finish
- 6) To prevent the formation of built-up –edge
- 7) To protect the work against the rusting
- 8) To improve surface finish,MRR

REQUIREMENTS OF CUTTING FLUID

- 1) Long life,
- 2) free from oxidation
- 3) High thermal conductivity
- 4) Low viscosity
- 5) Should be transparent
- 6) No fire or Accidental Hazards
- 7) Should have high flash point
- 8) Low price ,Easily available

4 categories of cutting fluids

- 1) Straight oils
- 2) Soluble oils
- 3) Semi-synthetic fluids
- 4) Synthetic fluids

1) Straight oils

- are the oldest class of engineered metal removal fluids.
- Used in machining operations in an undiluted form.
- They are composed of a base mineral or petroleum oil and often contains polar lubricants such as fats,

- Vegetable oils and esters as well as extreme pressure additives such as Chlorine, Sulphur and Phosphorus.
- Straight oils provide the best lubrication and the poorest cooling characteristics among cutting fluids.
- Chemically stable and lower in cost

2) Soluble Oil

- form an emulsion when mixed with water.
- The concentrate consists of a base mineral oil and emulsifiers to help produce a stable emulsion.
- They are used in a diluted form (usual concentration = 3 to 10%)
- Provide good lubrication and heat transfer performance.
- They are widely used in industry and are the least expensive among all cutting fluids.

3) Semi-synthetic fluids

- are essentially combination of synthetic and soluble oil fluids
- have characteristics common to both types.
- The cost and heat transfer performance of semi-synthetic fluids lie between those of synthetic and soluble oil fluids.

4) Synthetic Fluids

- contains no petroleum or mineral oil base
- are formulated from alkaline inorganic and organic compounds along with additives for corrosion inhibition.
- They are generally used in a diluted form (usual concentration = 3 to 10%).
- Synthetic fluids often provide the best cooling performance among all cutting fluids.

MACHINABILITY

The ease with which a given work material can be machined under a given set of cutting conditions.

The machinability of different materials can be compared in terms of the values of tool life, cutting forces and surface finish under similar cutting conditions.

The maximum production rate consistent with specified surface finish can serve as criteria for machinability.

Machinability index is a standardized value to express machinability of material.

The index is given as

$$\text{Machinability index, (\%)} = \frac{\text{Cutting speed of material for 20 minute tool life}}{\text{Cutting speed of standard steel for 20 minute tool life}} \times 100$$

FACTORS AFFECTING MACHINABILITY

a. MACHINE VARIABLES

1. Cutting speed
2. Feed and depth of cut
3. Tool geometry
4. Tool material
5. Cutting fluid
6. Rigidity of the system
7. Nature of cutting(Continuous or intermittent)

b. MATERIAL VARIABLES

1. Hardness
2. Tensile properties
3. Microstructure
4. Chemical composition
5. Degree of cold working
6. Strain hardenability

7. Shape and dimension of the work
8. Rigidity of workpiece

Tool life: If tool life is higher, then it indicates that the machinability of metal is higher.

Surface finish: Better surface finish obtained on the work piece indicates better machinability.

Power consumption: lower power consumption during the machining process indicates better machinability.

Cutting force: If cutting force required for machining is less, then it shows that the machinability of the metal is good

Hardness of metals: If the metal is very hard, then it will require more power consumption and it will generate high temperatures. Hence the tool wear may occur which leads to poor machinability.

Chemical composition: The cutting force and tool wear are very high while machining a pure metal. But if a small amount of carbon, manganese, lead, sulfur and phosphorus are added to the steels, it results in improved machinability.

Microstructure: If the metal contains abrasive inclusions and non-uniform structure with large and distorted grains, then the metals will show less machinability.

Treatment given to metal: Cold worked steels show better machinability as it increases tool life as well as its machining can be done at high cutting speed. Machinability of high carbon steels can be increased by their hot working.
